US012412166B2

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Kawata et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **MOBILE SALES SYSTEM AND SERVER DEVICE**

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(73) Assignee: **Toshiba Tec Kabushiki Kaisha**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 232 days.

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(22) Filed: **Jan. 20, 2023**

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(30) **Foreign Application Priority Data**

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G06Q 20/32 (2012.01)
G06Q 20/20 (2012.01)
G06Q 30/0601 (2023.01)

(52) **U.S. Cl.**
CPC ***G06Q 20/3224*** (2013.01); ***G06Q 20/206*** (2013.01); ***G06Q 20/3223*** (2013.01); ***G06Q 30/0639*** (2013.01)

(58) **Field of Classification Search**
CPC G06Q 20/3224; G06Q 20/206; G06Q 20/3223; G06Q 30/0639; G06Q 20/18;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,216,188 B2 * 2/2019 Brady G06Q 10/0837
10,636,019 B1 * 4/2020 Abrons G06Q 30/0207
(Continued)

FOREIGN PATENT DOCUMENTS

CN 101069219 A * 11/2007 .... G07F 11/002
CN 108629896 A * 10/2018 .... G06Q 20/18
(Continued)

OTHER PUBLICATIONS

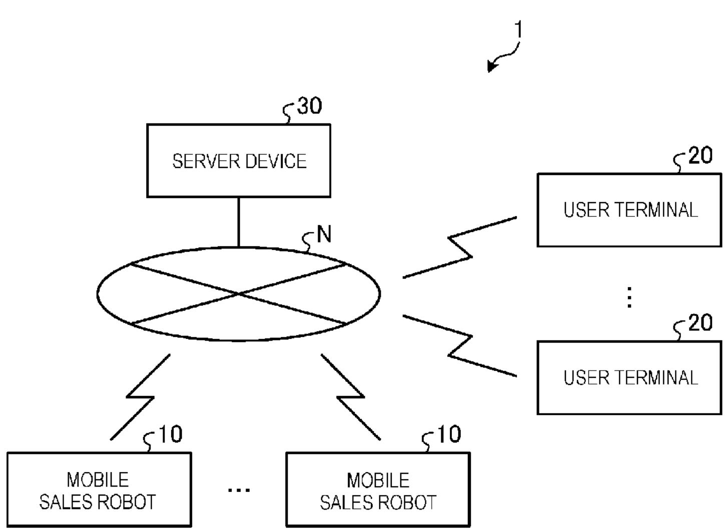
Japanese Office Action dated May 27, 2025, mailed in counterpart Japanese Application No. 2022-025950, 8 pages.

Primary Examiner — Ashford S Hayles
(74) *Attorney, Agent, or Firm* — Kim & Stewart LLP

(57) **ABSTRACT**

According to one embodiment, a mobile sales system includes a terminal device and a mobile sales device on which items to be sold can be mounted. The terminal device is used to designate a customer call position for the mobile sales device and is configured to transmit position information indicating the customer call position. The mobile sales device is configured to acquire the position information transmitted by the terminal device, move to a location corresponding to the acquired position information, and execute payment processing for an item removed from the mobile sales device by a customer.

20 Claims, 24 Drawing Sheets



(58) **Field of Classification Search**
CPC G06Q 20/20; G06Q 20/202; G06Q 20/322; G06Q 20/40145; G06Q 30/06; G06Q 30/0601; G07F 9/001; G07F 9/002; G07F 11/00; G07F 11/04; G07G 1/0045
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,748,143	B2 *	8/2020	Edgington	G06Q 20/405
11,270,257	B2	3/2022	Nakagawa	
11,392,864	B2	7/2022	Qiu	
11,488,212	B2	11/2022	Kobayashi	
11,501,523	B2	11/2022	Li	
11,556,976	B2	1/2023	Kobayashi	
D1,039,580	S *	8/2024	Huang	D15/199
2008/0306826	A1 *	12/2008	Kramer	G06Q 30/0226 705/14.27
2009/0248574	A1 *	10/2009	Leung	G06Q 20/10 705/39
2014/0012622	A1 *	1/2014	Kamiya	G06Q 30/0201 705/7.29
2014/0136414	A1 *	5/2014	Abhyanker	G06Q 20/3224 701/25
2014/0143061	A1 *	5/2014	Abhyanker	G06Q 50/01 705/14.58
2016/0232489	A1 *	8/2016	Skaaksrud	G06Q 10/0836
2017/0011580	A1 *	1/2017	Huang	G06Q 30/0631
2017/0018041	A1 *	1/2017	Fox	G06Q 20/3224
2017/0132934	A1 *	5/2017	Kentley	G05D 1/0088
2017/0200197	A1 *	7/2017	Brubaker	B60Q 1/2619
2018/0053369	A1 *	2/2018	High	G07F 9/0235
2018/0099602	A1 *	4/2018	Salter	B60Q 1/507
2018/0164818	A1 *	6/2018	Wilkinson	G06V 20/584
2018/0215543	A1 *	8/2018	Lert, Jr.	G06Q 10/08
2018/0246513	A1 *	8/2018	Cronin	G07F 9/001
2018/0259973	A1 *	9/2018	Maruo	G05D 1/0011
2018/0260778	A1 *	9/2018	Mazetti	G06Q 10/0875
2018/0267963	A1 *	9/2018	Ogata	G06Q 30/0269
2018/0349872	A1 *	12/2018	Ahmed	G07G 1/0036
2019/0034858	A1 *	1/2019	Ferguson	G05D 1/6445
2019/0035044	A1 *	1/2019	Ferguson	G05D 1/6445
2019/0049988	A1 *	2/2019	Meij	G06Q 50/12
2019/0051083	A1 *	2/2019	Goldberg	G07F 9/026
2019/0051090	A1 *	2/2019	Goldberg	B62D 63/04
2019/0236534	A1 *	8/2019	Kaneko	G05D 1/0274
2020/0105085	A1 *	4/2020	Jafa	G08G 1/202
2020/0209821	A1 *	7/2020	Heinla	G05B 19/4155
2020/0301433	A1 *	9/2020	Katayama	G06Q 20/145
2020/0364655	A1 *	11/2020	Burris	G07F 7/025
2021/0094506	A1 *	4/2021	Baker	B60R 25/241
2021/0150511	A1 *	5/2021	Mandloi	G06Q 20/40145
2021/0295635	A1 *	9/2021	Matsunami	H04W 4/40
2021/0302971	A1 *	9/2021	Cousins	G05D 1/0094
2021/0303105	A1 *	9/2021	Yoneyama	G06Q 10/06312
2021/0330119	A1 *	10/2021	Väin	G07F 13/065
2022/0004974	A1 *	1/2022	Heinla	G05D 1/43
2022/0012677	A1 *	1/2022	Rongley	G06K 7/10297
2022/0180299	A1 *	6/2022	Väin	G06Q 10/087
2022/0207505	A1 *	6/2022	Ahmed	G06Q 20/0855
2022/0266738	A1 *	8/2022	Meador	E05B 49/00
2022/0415114	A1 *	12/2022	Long, II	G07F 9/001
2023/0144456	A1 *	5/2023	Ferguson	G06Q 20/401 705/26.81
2023/0153886	A1 *	5/2023	Arai	G06Q 30/0202 705/26.7
2023/0184557	A1 *	6/2023	Meador	B60P 3/007 705/330
2023/0186368	A1 *	6/2023	Arai	G06Q 30/0261 705/26.7
2024/0336183	A1 *	10/2024	Bolshchikov	G07F 9/001

FOREIGN PATENT DOCUMENTS

CN	109035564	A	*	12/2018	G07F 9/002
CN	109670758	A	*	4/2019	G06Q 10/083
CN	110084961	A	*	8/2019	
CN	114202328	A	*	3/2022	
CN	116524649	A	*	8/2023	
JP	2004102694	A	*	4/2004	
JP	2004295377	A		10/2004	
JP	2008087892	A		4/2008	
JP	2008123278	A		5/2008	
JP	2011221984	A	*	11/2011	
JP	2015069594	A		4/2015	
JP	2020126559	A		8/2020	
JP	2021507203	A		2/2021	
JP	2021033471	A		3/2021	
JP	2021039620	A		3/2021	
KR	20210105527	A	*	8/2021	
KR	102406988	B1	*	6/2022	
WO	2018163547	A1		9/2018	
WO	WO-2018191504	A1	*	10/2018	B60Q 1/50
WO	WO-2021100096	A1	*	5/2021	G02B 27/017
WO	WO-2024248205	A1	*	12/2024	G06Q 30/0213

* cited by examiner

FIG. 1

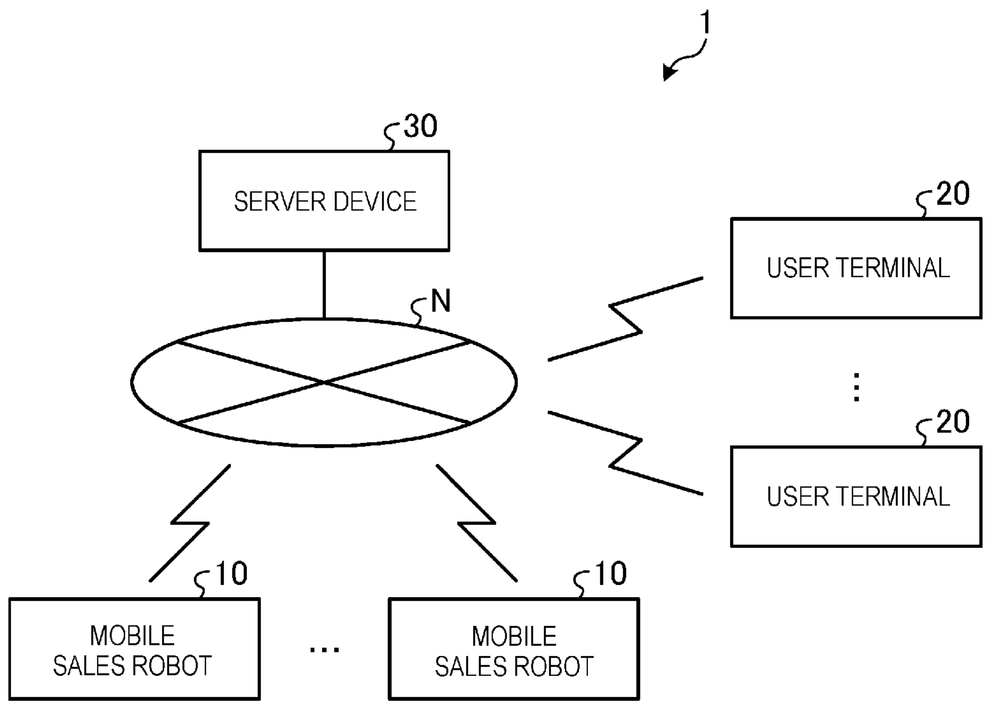
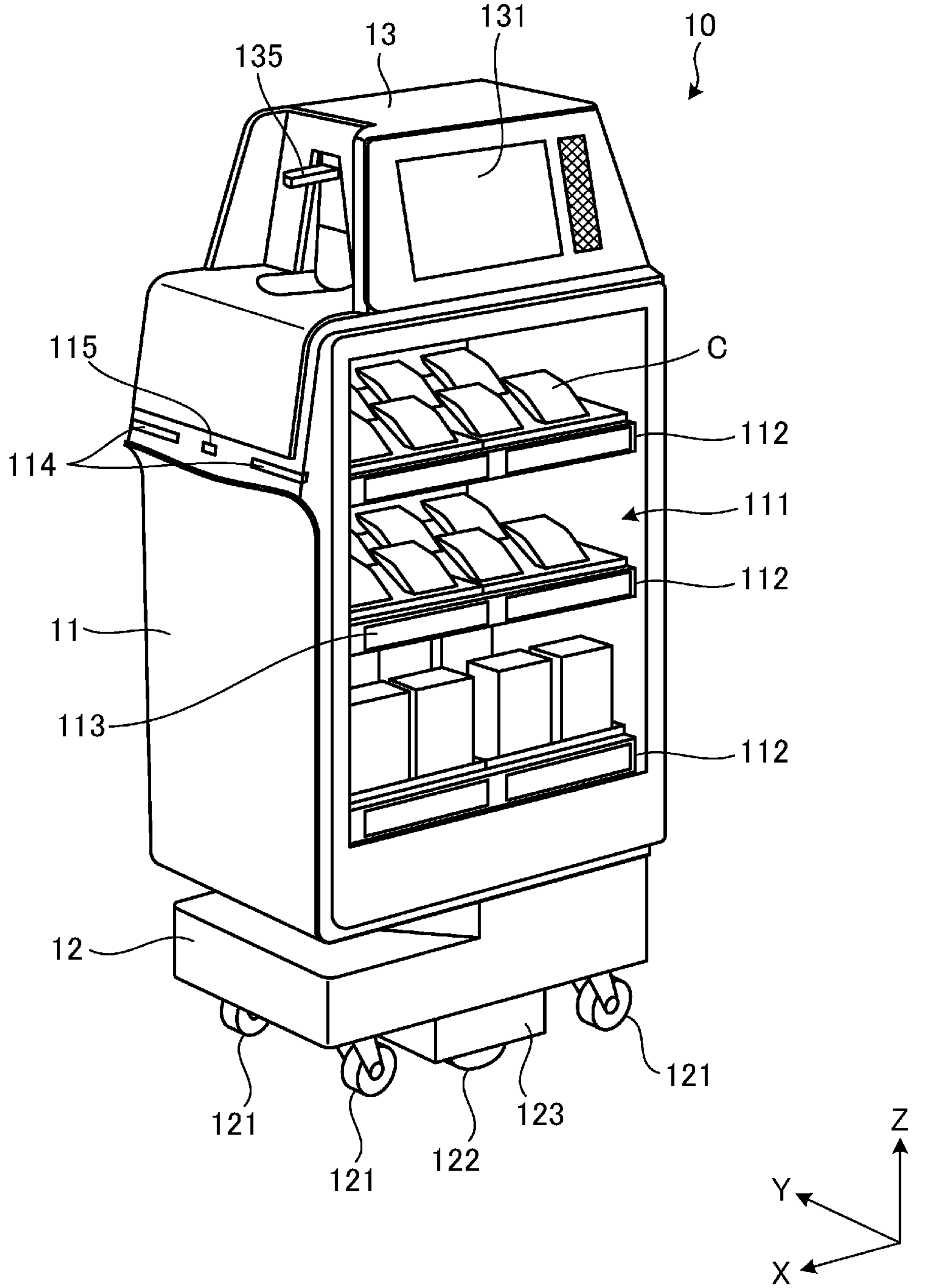

1
30
SERVER DEVICE
20
USER TERMINAL
N
20
USER TERMINAL
10
MOBILE SALES ROBOT
10
MOBILE SALES ROBOT

FIG. 2



10
13
131
135
115
114
C
112
111
112
11
113
112
12
121
121
121
122
123
121
Z
Y
X

FIG. 3

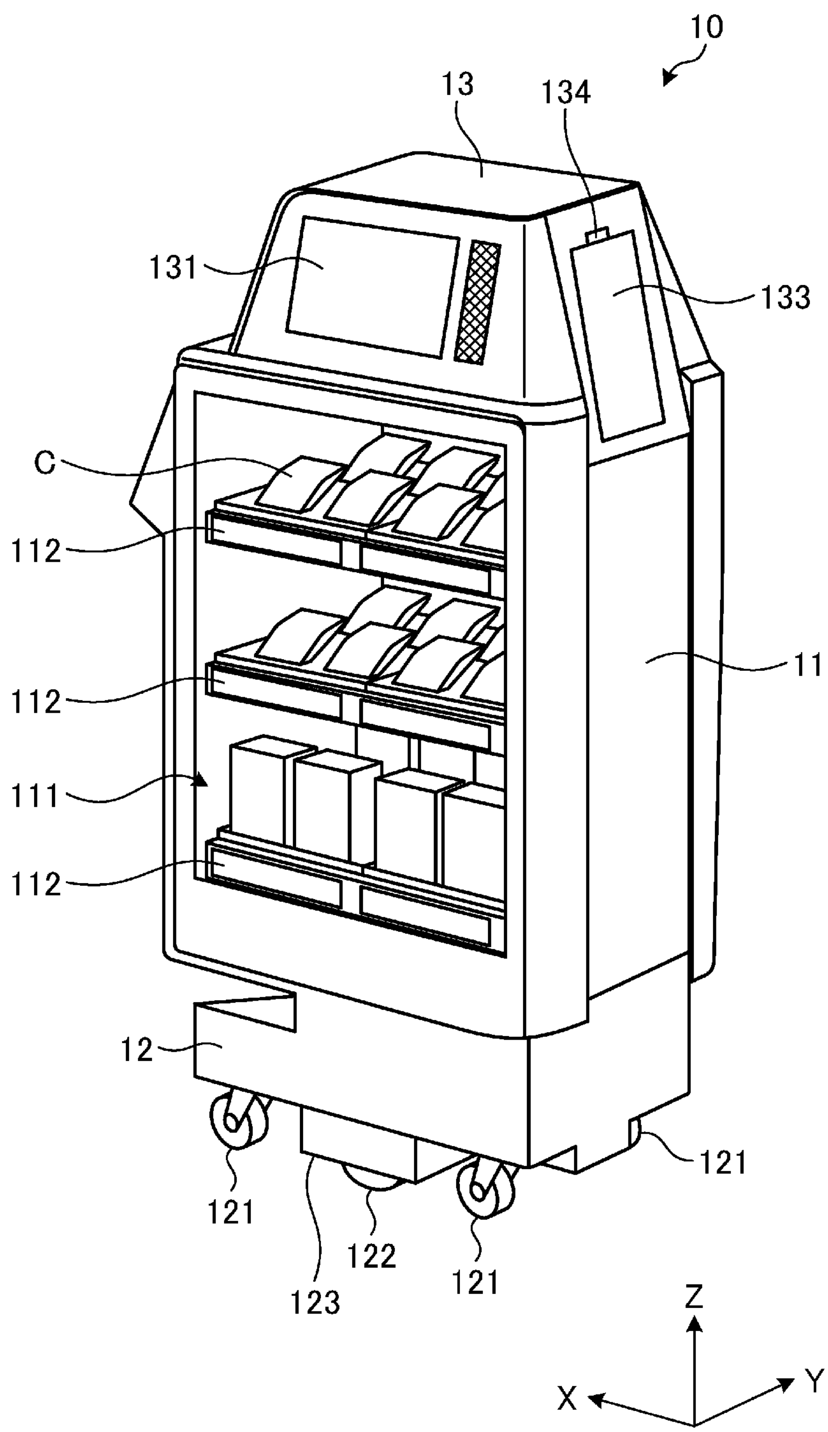
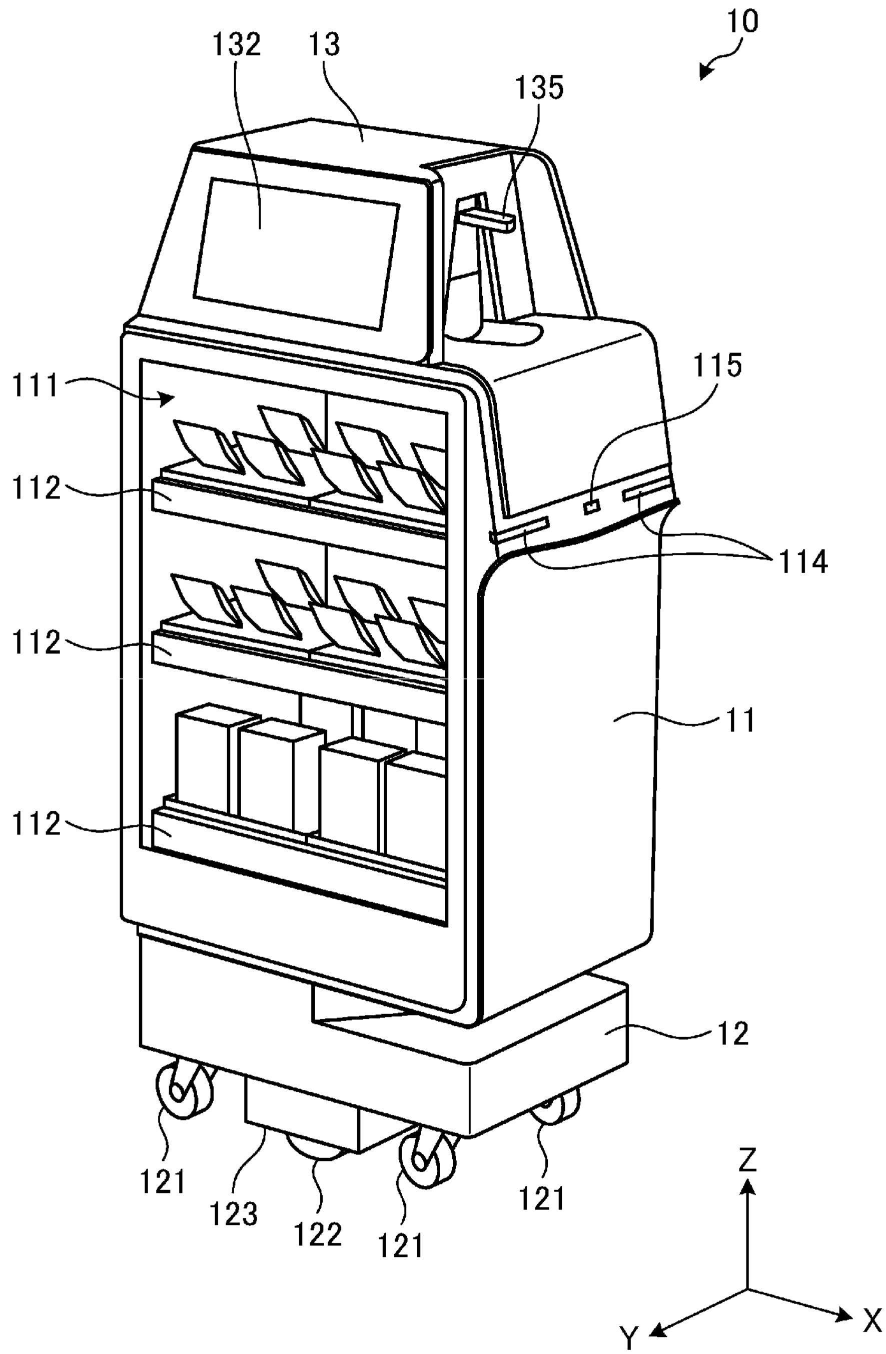


FIG. 4



10
132
13
135
111
115
112
114
112
11
112
12
121
123
122
121
121
Z
Y
X

FIG. 5

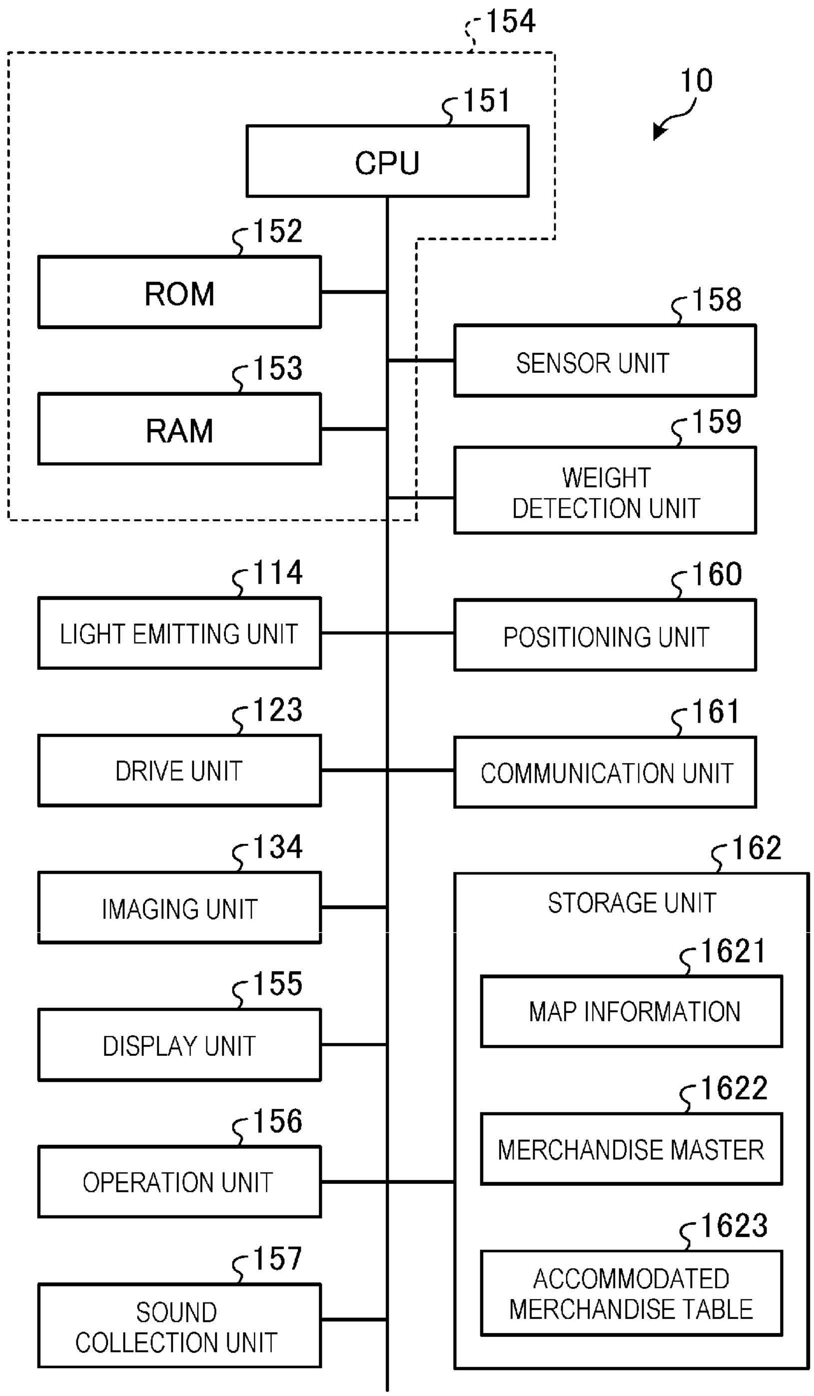

154
10
151
CPU
152
ROM
153
RAM
158
SENSOR UNIT
159
WEIGHT DETECTION UNIT
114
LIGHT EMITTING UNIT
160
POSITIONING UNIT
123
DRIVE UNIT
161
COMMUNICATION UNIT
134
IMAGING UNIT
162
STORAGE UNIT
1621
MAP INFORMATION
155
DISPLAY UNIT
1622
MERCHANDISE MASTER
156
OPERATION UNIT
1623
ACCOMMODATED MERCHANDISE TABLE
157
SOUND COLLECTION UNIT

FIG. 6

1622

MERCHANDISE CODE	MERCHANDISE INFORMATION					
	MERCHANDISE NAME	TYPE	PRICE	WEIGHT	IMAGE DATA	...

FIG. 7

1623

COLUMN ID	MERCHANDISE CODE	QUANTITY	...

FIG. 8

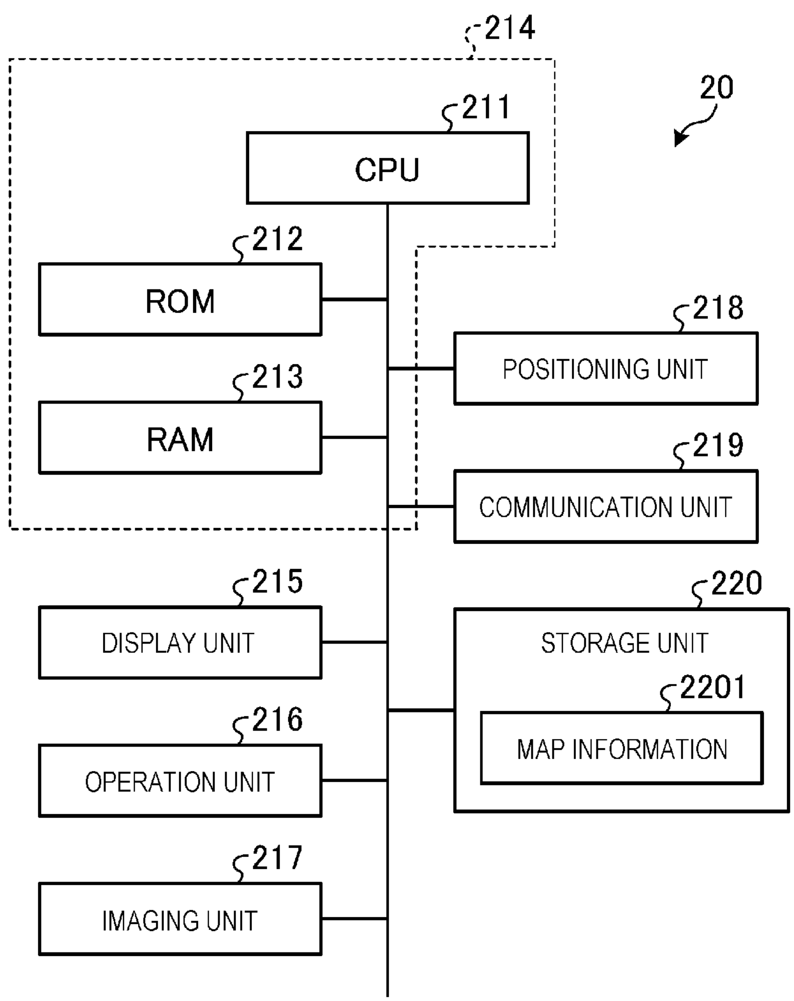

214
20
211
CPU
212
ROM
213
RAM
218
POSITIONING UNIT
219
COMMUNICATION UNIT
215
DISPLAY UNIT
220
STORAGE UNIT
2201
MAP INFORMATION
216
OPERATION UNIT
217
IMAGING UNIT

FIG. 9

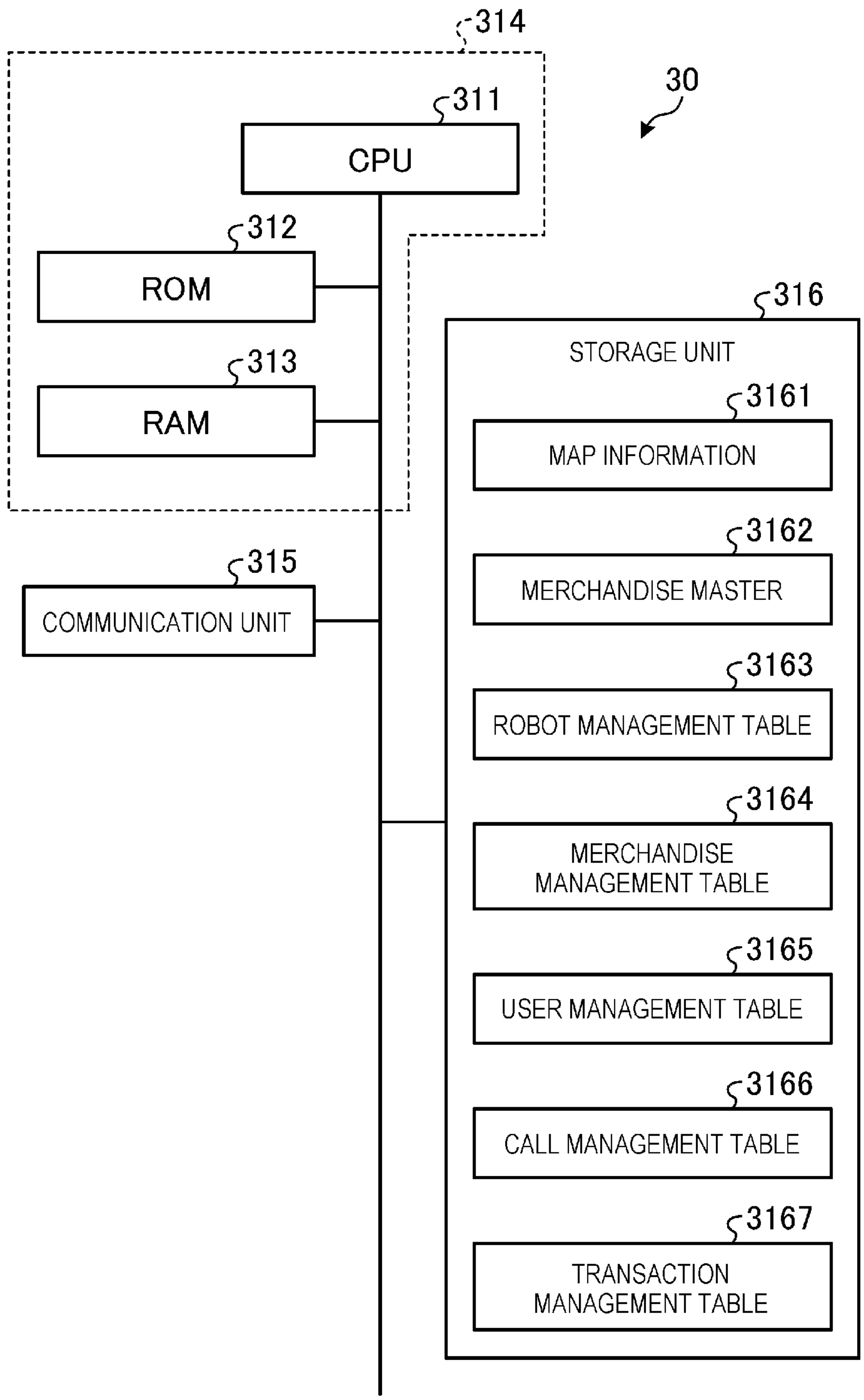

314
30
311
CPU
312
ROM
313
RAM
315
COMMUNICATION UNIT
316
STORAGE UNIT
3161
MAP INFORMATION
3162
MERCHANDISE MASTER
3163
ROBOT MANAGEMENT TABLE
3164
MERCHANDISE MANAGEMENT TABLE
3165
USER MANAGEMENT TABLE
3166
CALL MANAGEMENT TABLE
3167
TRANSACTION MANAGEMENT TABLE

FIG. 10

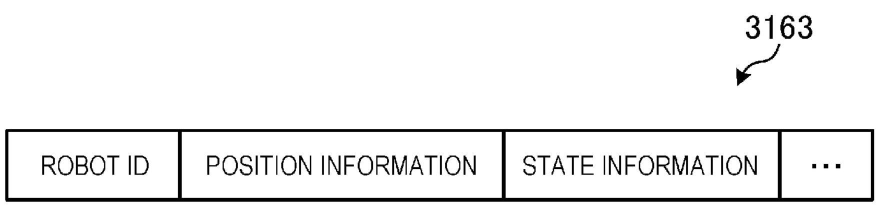

3163
ROBOT ID
POSITION INFORMATION
STATE INFORMATION
...

FIG. 11

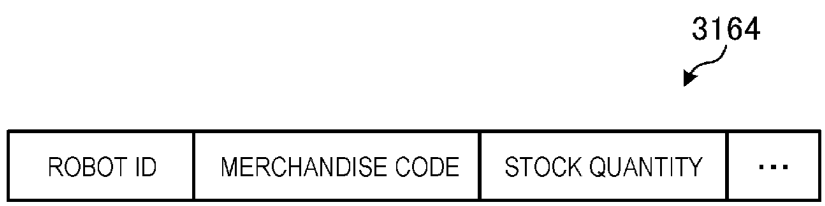

3164
ROBOT ID
MERCHANDISE CODE
STOCK QUANTITY
...

FIG. 12

3165

USER ID	USER INFORMATION		
	FEATURE INFORMATION	SETTLEMENT INFORMATION	...

FIG. 13

3166

TERMINAL ID	CALL DATE AND TIME	CALL POSITION	AUTHENTICATION INFORMATION	ROBOT ID	ARRIVAL DATE AND TIME	...

FIG. 14

3167

TRANSACTION ID	ROBOT ID	USER ID	MERCHANDISE CODE	SETTLEMENT FLAG	...

FIG. 15

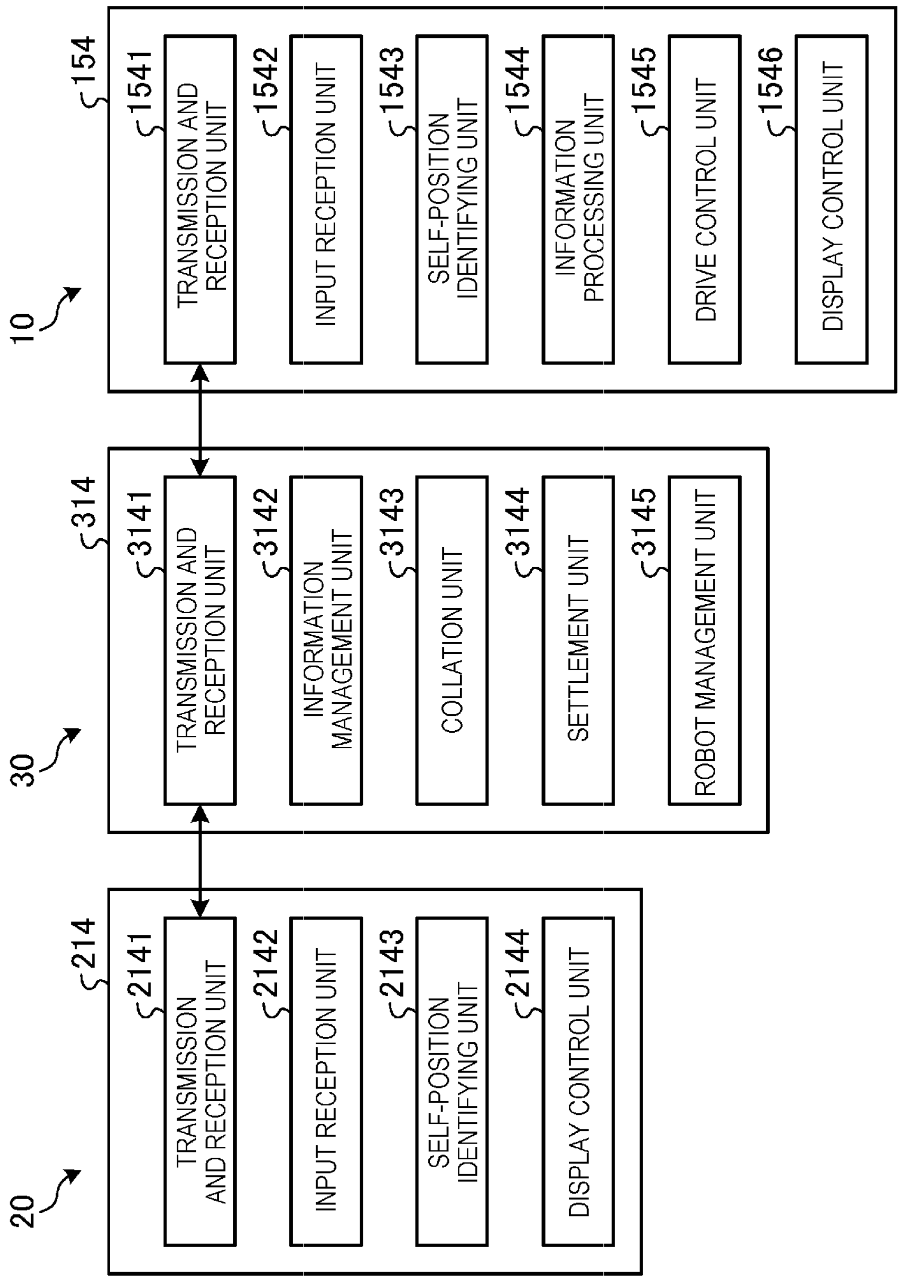

10
154
1541
TRANSMISSION AND RECEPTION UNIT
1542
INPUT RECEPTION UNIT
1543
SELF-POSITION IDENTIFYING UNIT
1544
INFORMATION PROCESSING UNIT
1545
DRIVE CONTROL UNIT
1546
DISPLAY CONTROL UNIT
30
314
3141
TRANSMISSION AND RECEPTION UNIT
3142
INFORMATION MANAGEMENT UNIT
3143
COLLATION UNIT
3144
SETTLEMENT UNIT
3145
ROBOT MANAGEMENT UNIT
20
214
2141
TRANSMISSION AND RECEPTION UNIT
2142
INPUT RECEPTION UNIT
2143
SELF-POSITION IDENTIFYING UNIT
2144
DISPLAY CONTROL UNIT

FIG. 16

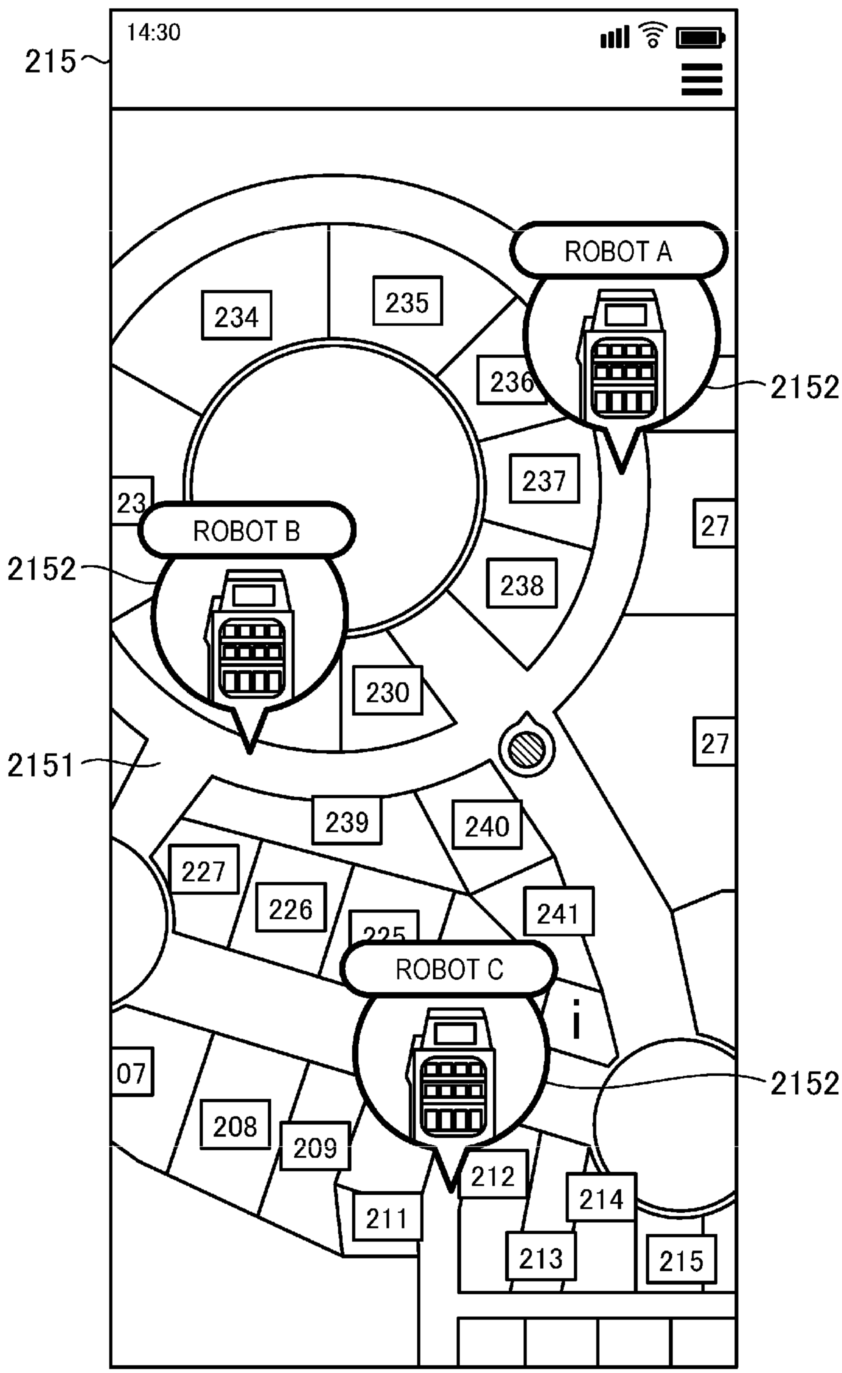

215
14:30
ROBOT A
ROBOT B
ROBOT C
2152
2152
2152
2151
234
235
236
237
238
230
239
240
241
227
226
225
208
209
211
212
213
214
215
23
27
27
07
i

FIG. 17

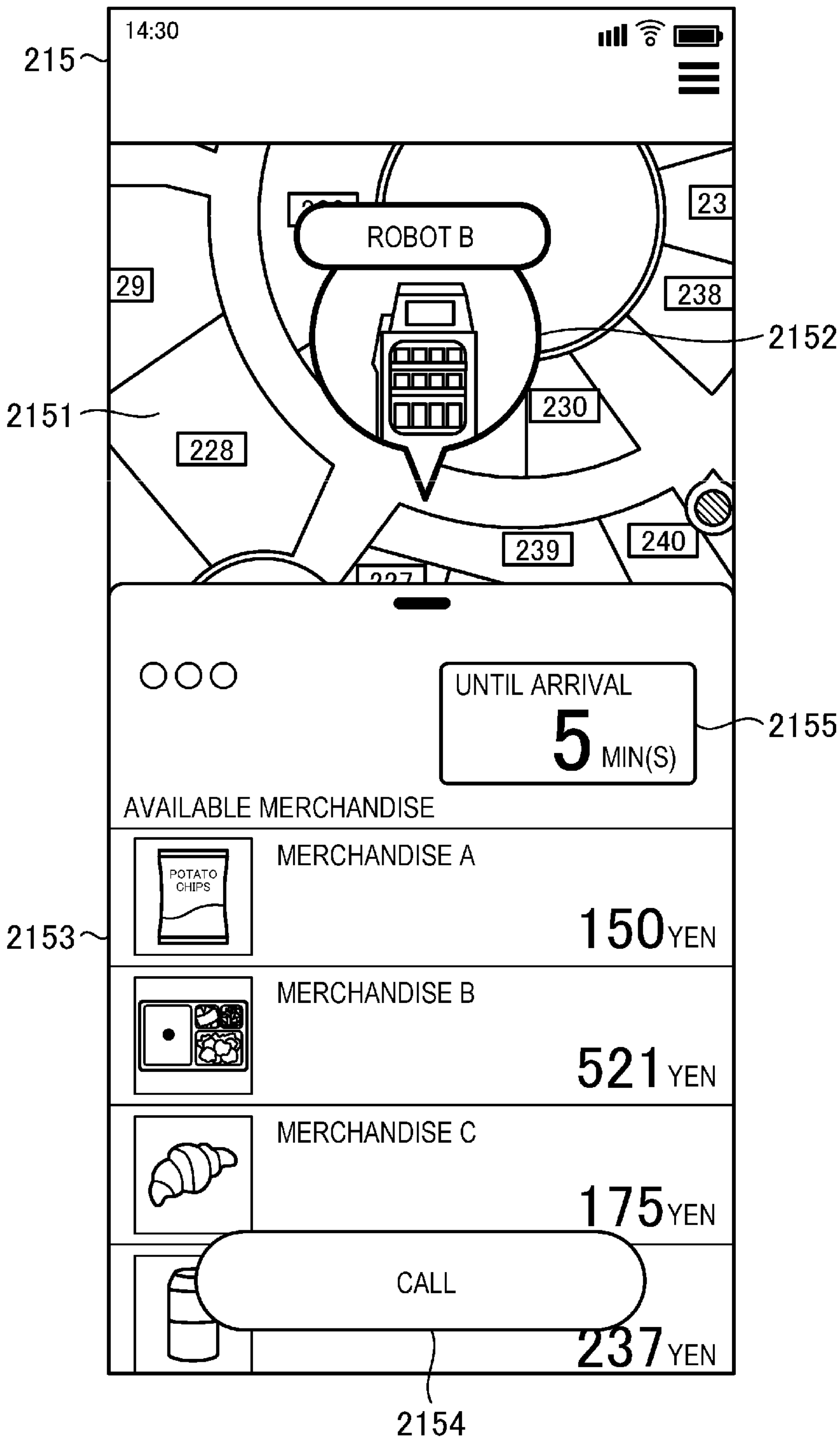

215
14:30
ROBOT B
23
29
238
2152
2151
230
228
239
240
○○○
UNTIL ARRIVAL
5 MIN(S)
2155
AVAILABLE MERCHANDISE
POTATO CHIPS
MERCHANDISE A
150YEN
2153
MERCHANDISE B
521YEN
MERCHANDISE C
175YEN
CALL
237YEN
2154

FIG. 18

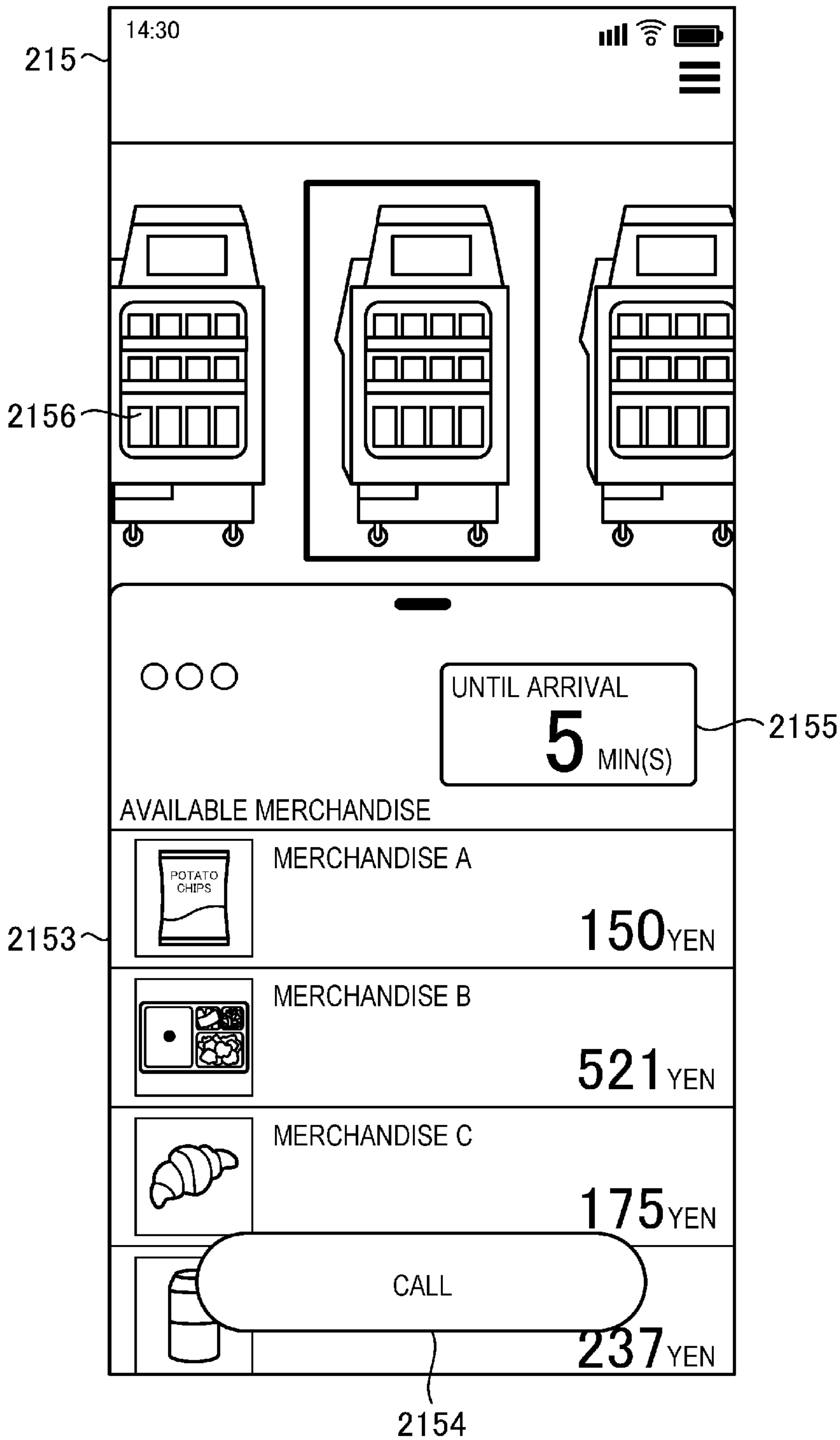
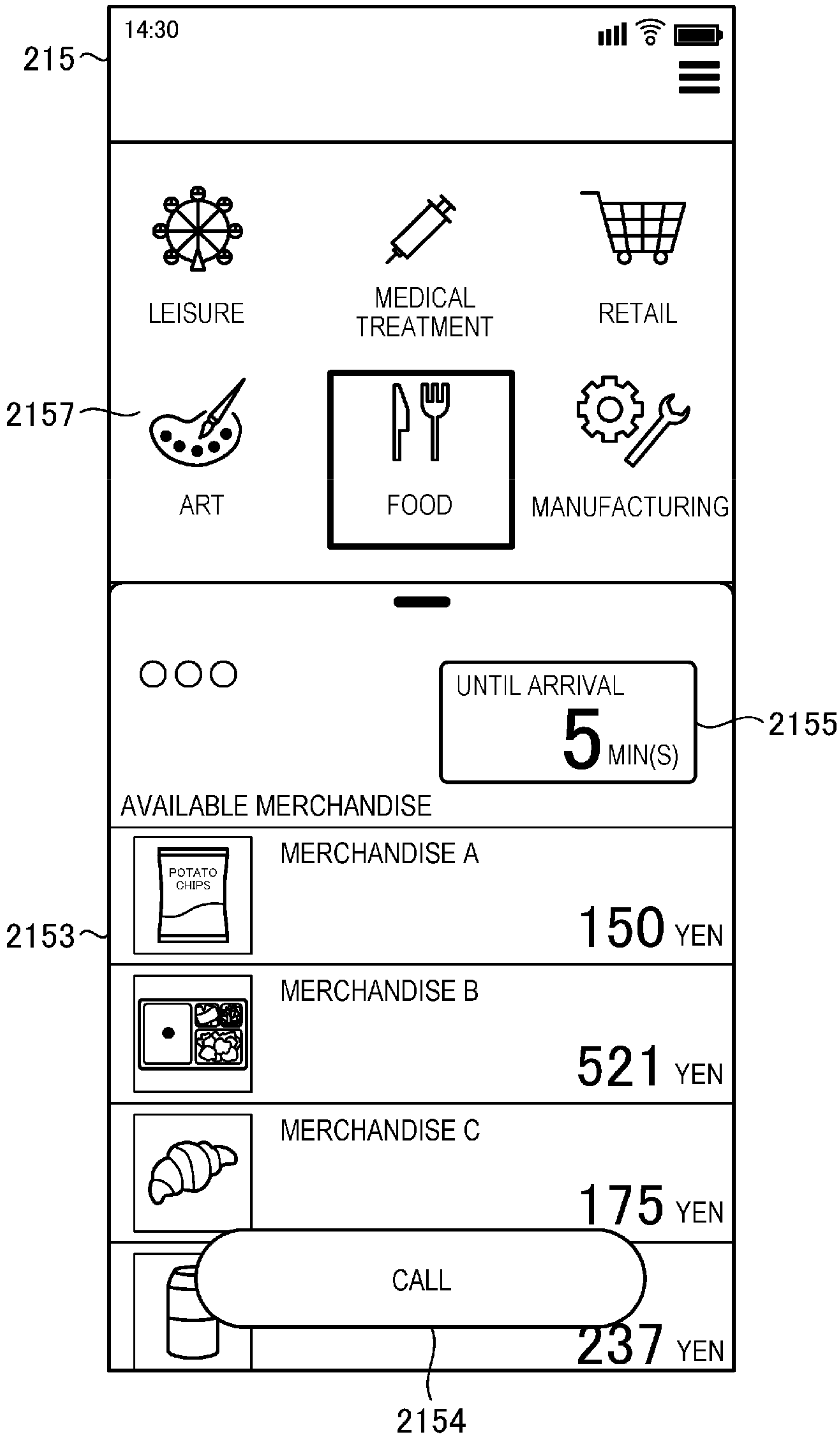

215
14:30
2156
○○○
UNTIL ARRIVAL
5 MIN(S)
2155
AVAILABLE MERCHANDISE
2153
POTATO CHIPS
MERCHANDISE A
150 YEN
MERCHANDISE B
521 YEN
MERCHANDISE C
175 YEN
CALL
237 YEN
2154

FIG. 19



215
14:30
LEISURE
MEDICAL TREATMENT
RETAIL
2157
ART
FOOD
MANUFACTURING
○○○
UNTIL ARRIVAL
5 MIN(S)
2155
AVAILABLE MERCHANDISE
2153
POTATO CHIPS
MERCHANDISE A
150 YEN
MERCHANDISE B
521 YEN
MERCHANDISE C
175 YEN
CALL
237 YEN
2154

FIG. 20

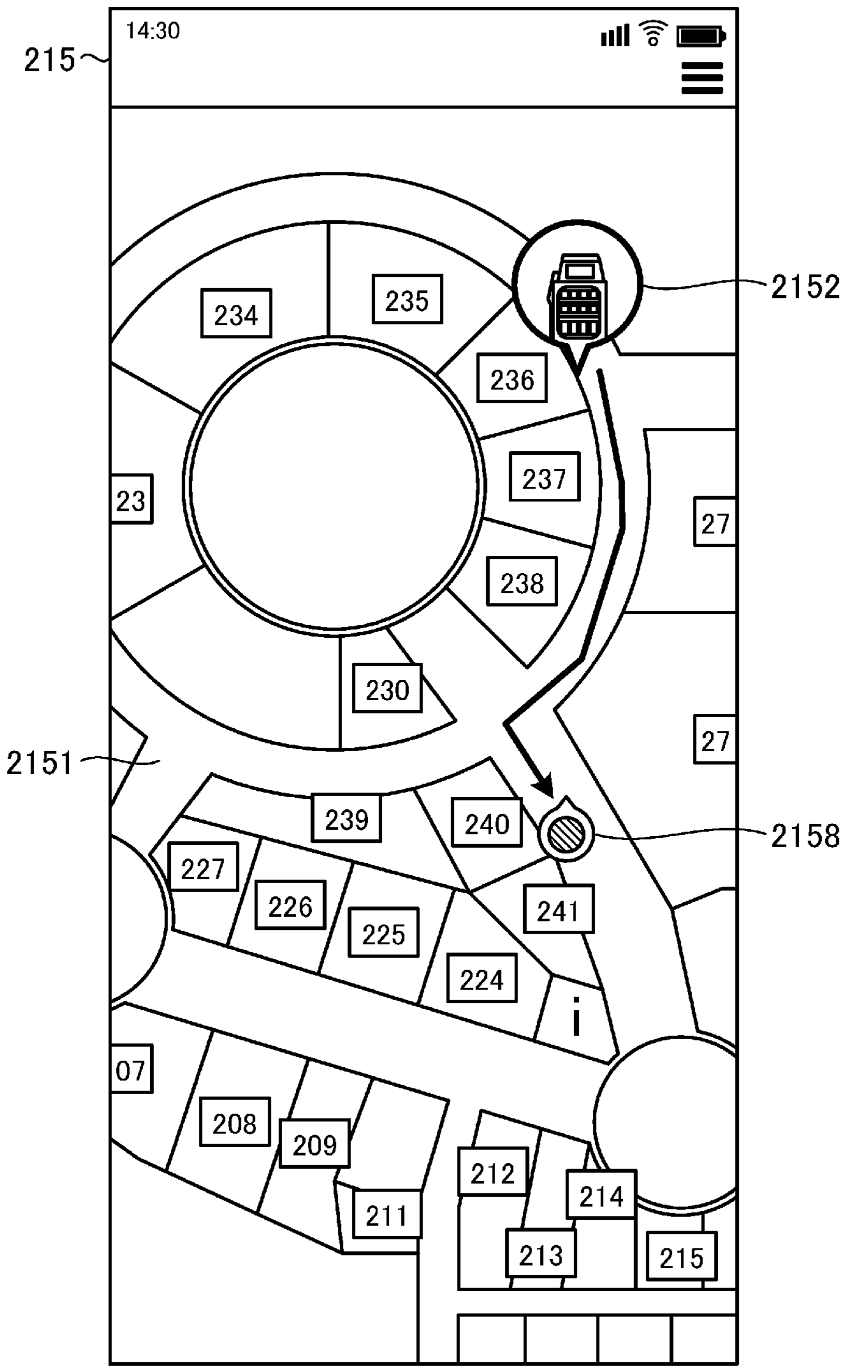

215
14:30
2152
234
235
236
237
238
23
27
230
2151
27
239
240
2158
227
226
225
241
224
i
07
208
209
212
214
211
213
215

FIG. 21

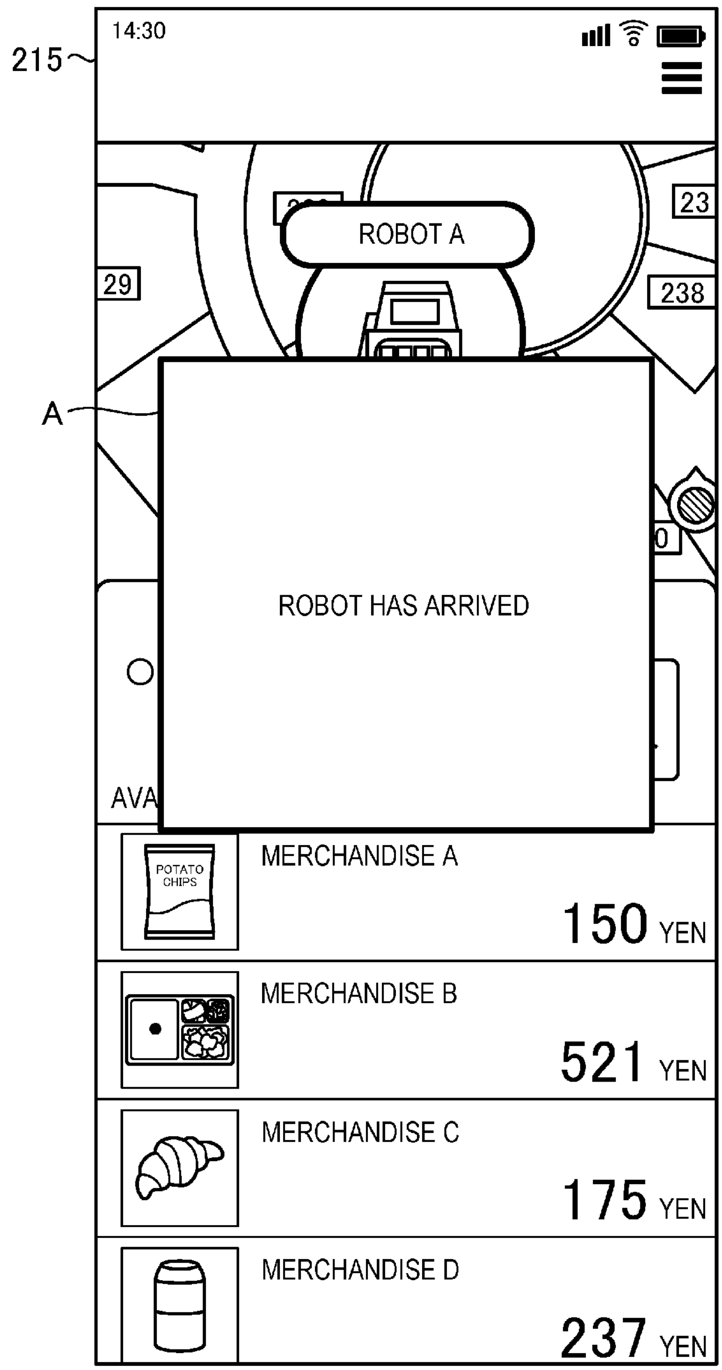

215
14:30
23
ROBOT A
29
238
A
0
ROBOT HAS ARRIVED
AVA
POTATO CHIPS
MERCHANDISE A
150 YEN
MERCHANDISE B
521 YEN
MERCHANDISE C
175 YEN
MERCHANDISE D
237 YEN

FIG. 22

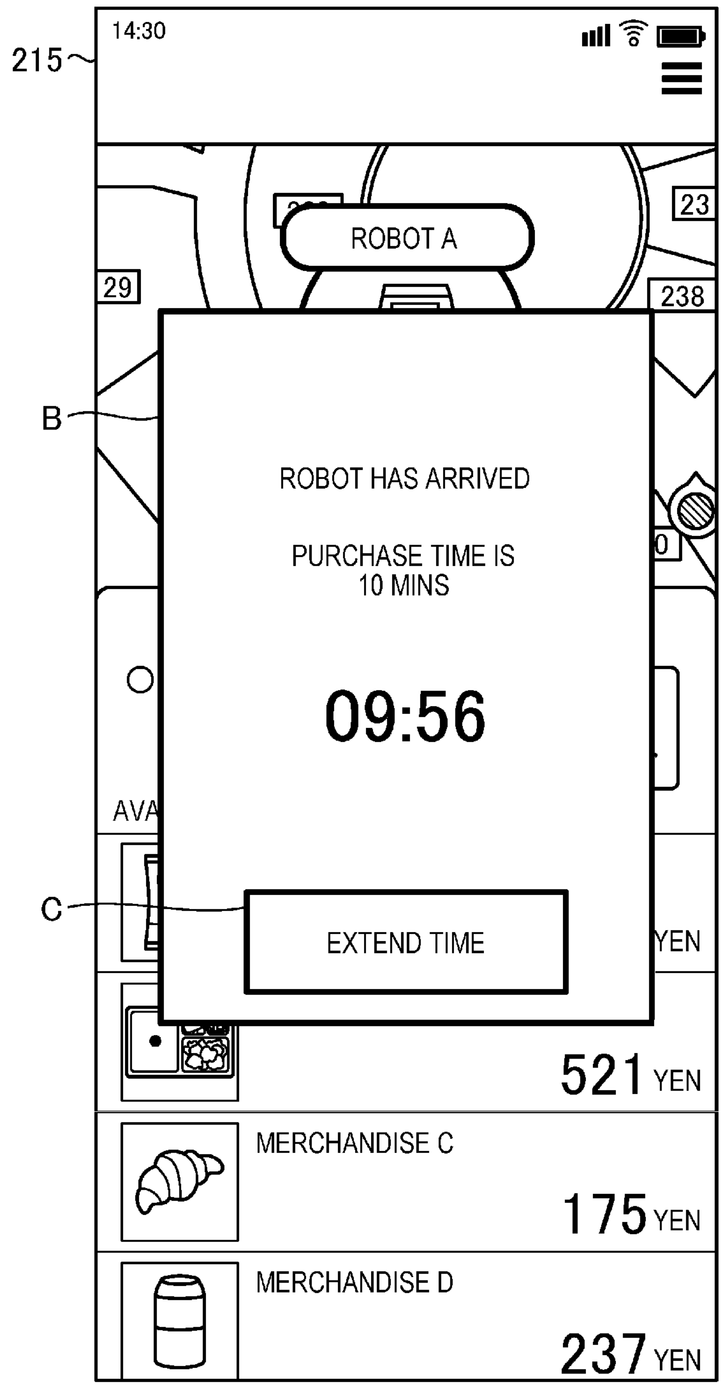

215
14:30
ROBOT A
23
29
238
B
ROBOT HAS ARRIVED
PURCHASE TIME IS
10 MINS
09:56
C
EXTEND TIME
YEN
521 YEN
MERCHANDISE C
175 YEN
MERCHANDISE D
237 YEN

FIG. 23

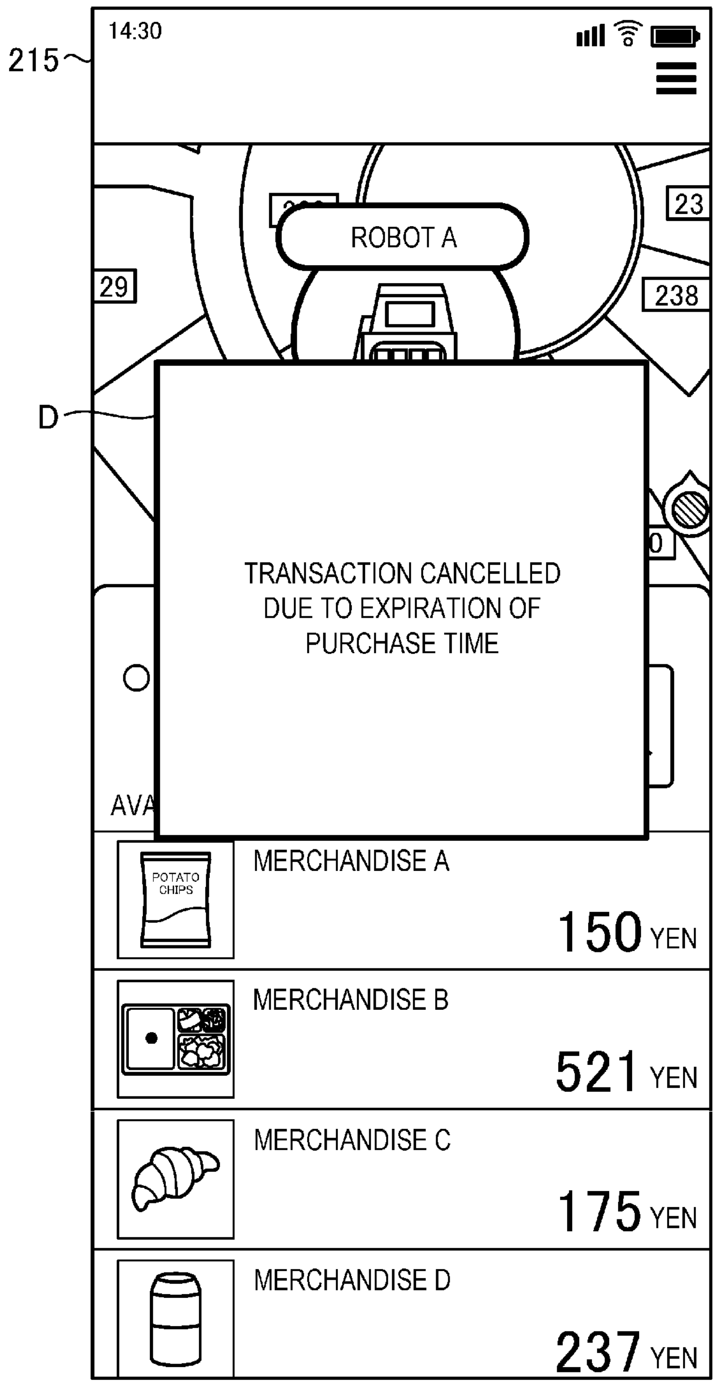

215
14:30
ROBOT A
23
29
238
D
0
TRANSACTION CANCELLED
DUE TO EXPIRATION OF
PURCHASE TIME
AVA
POTATO CHIPS
MERCHANDISE A
150 YEN
MERCHANDISE B
521 YEN
MERCHANDISE C
175 YEN
MERCHANDISE D
237 YEN

FIG. 24

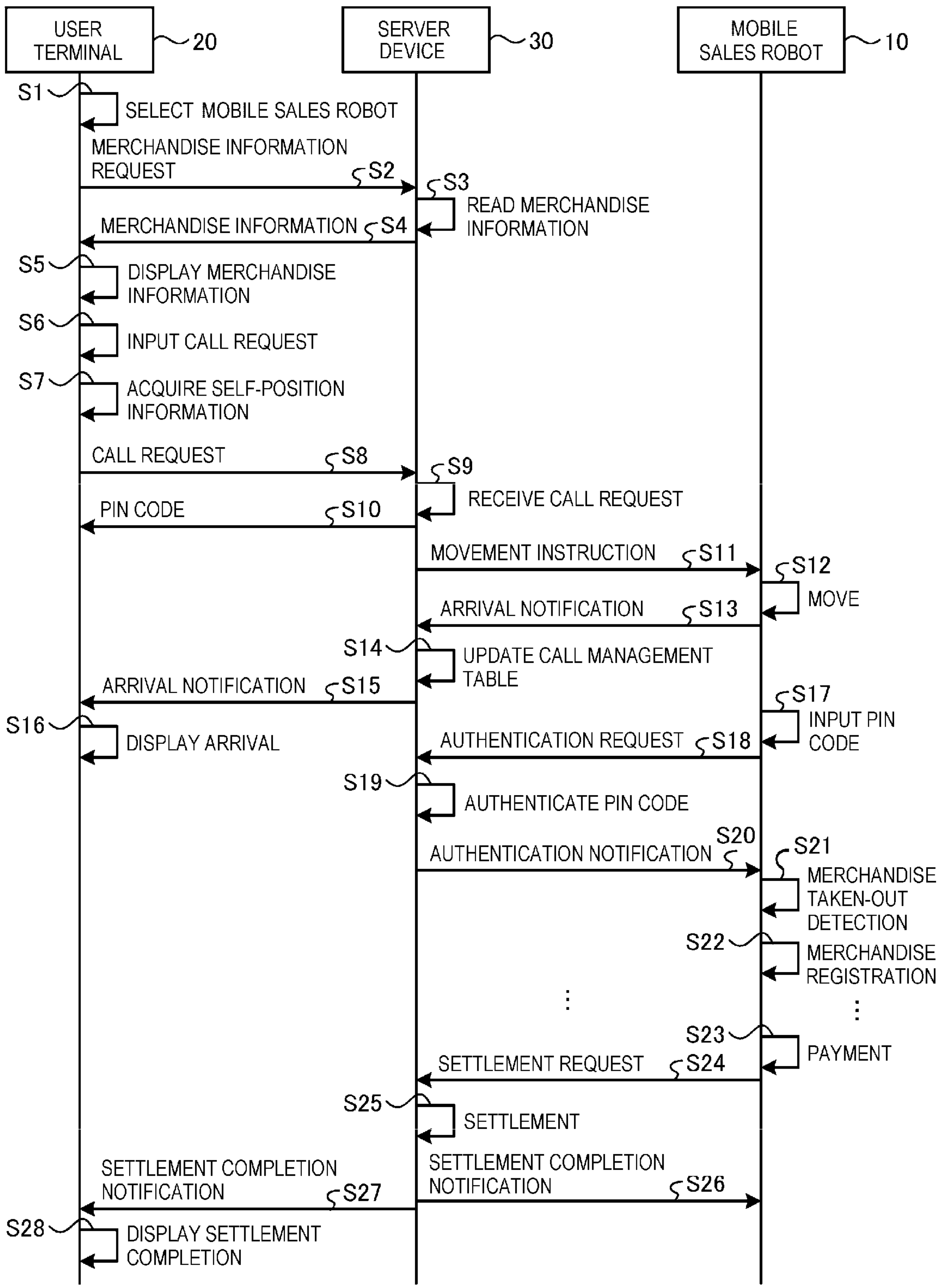

USER TERMINAL
20
SERVER DEVICE
30
MOBILE SALES ROBOT
10
S1
SELECT MOBILE SALES ROBOT
MERCHANDISE INFORMATION REQUEST
S2
S3
READ MERCHANDISE INFORMATION
MERCHANDISE INFORMATION
S4
S5
DISPLAY MERCHANDISE INFORMATION
S6
INPUT CALL REQUEST
S7
ACQUIRE SELF-POSITION INFORMATION
CALL REQUEST
S8
S9
RECEIVE CALL REQUEST
PIN CODE
S10
MOVEMENT INSTRUCTION
S11
S12
MOVE
ARRIVAL NOTIFICATION
S13
S14
UPDATE CALL MANAGEMENT TABLE
ARRIVAL NOTIFICATION
S15
S16
DISPLAY ARRIVAL
S17
INPUT PIN CODE
AUTHENTICATION REQUEST
S18
S19
AUTHENTICATE PIN CODE
AUTHENTICATION NOTIFICATION
S20
S21
MERCHANDISE TAKEN-OUT DETECTION
S22
MERCHANDISE REGISTRATION
S23
PAYMENT
SETTLEMENT REQUEST
S24
S25
SETTLEMENT
SETTLEMENT COMPLETION NOTIFICATION
S27
SETTLEMENT COMPLETION NOTIFICATION
S26
S28
DISPLAY SETTLEMENT COMPLETION

FIG. 25

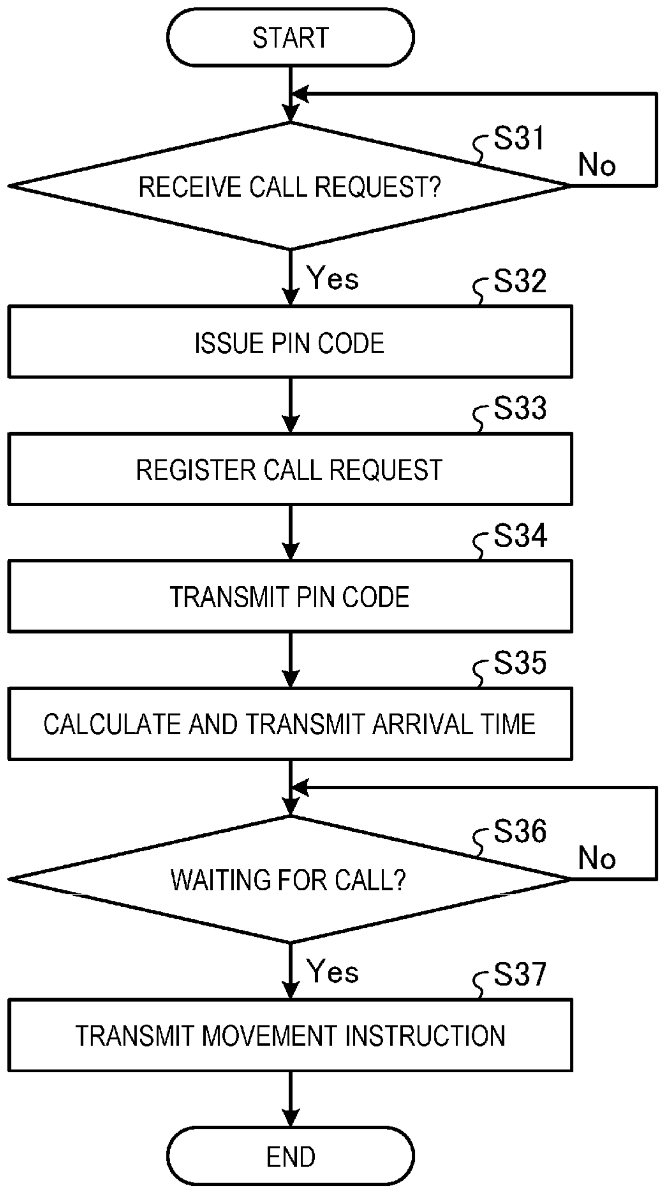

START
S31
RECEIVE CALL REQUEST?
No
Yes
S32
ISSUE PIN CODE
S33
REGISTER CALL REQUEST
S34
TRANSMIT PIN CODE
S35
CALCULATE AND TRANSMIT ARRIVAL TIME
S36
WAITING FOR CALL?
No
Yes
S37
TRANSMIT MOVEMENT INSTRUCTION
END

FIG. 26

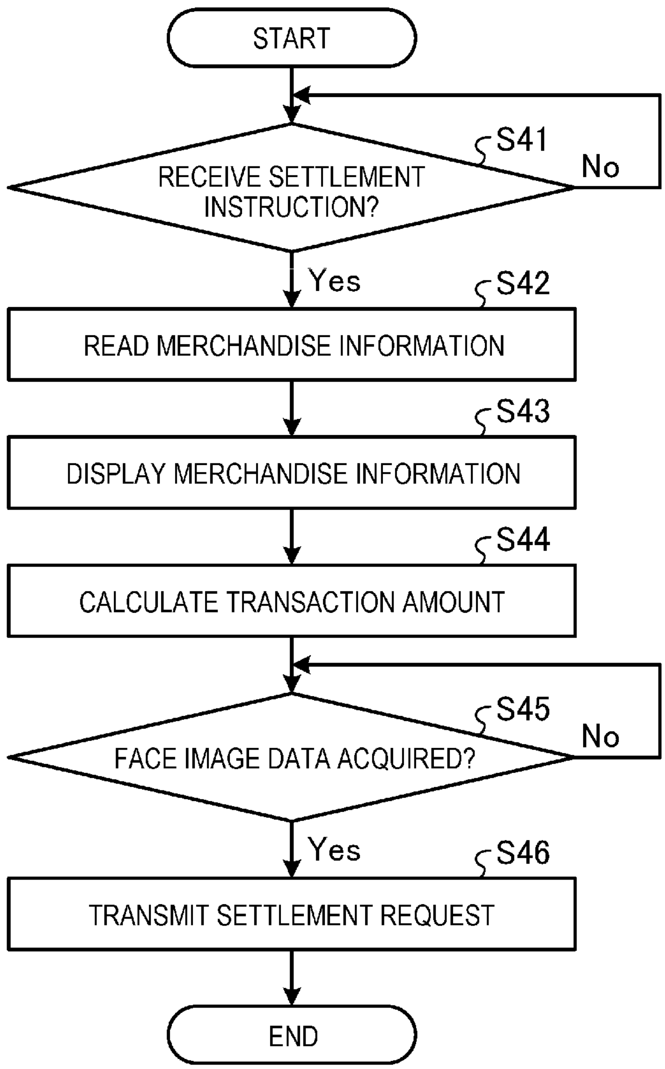
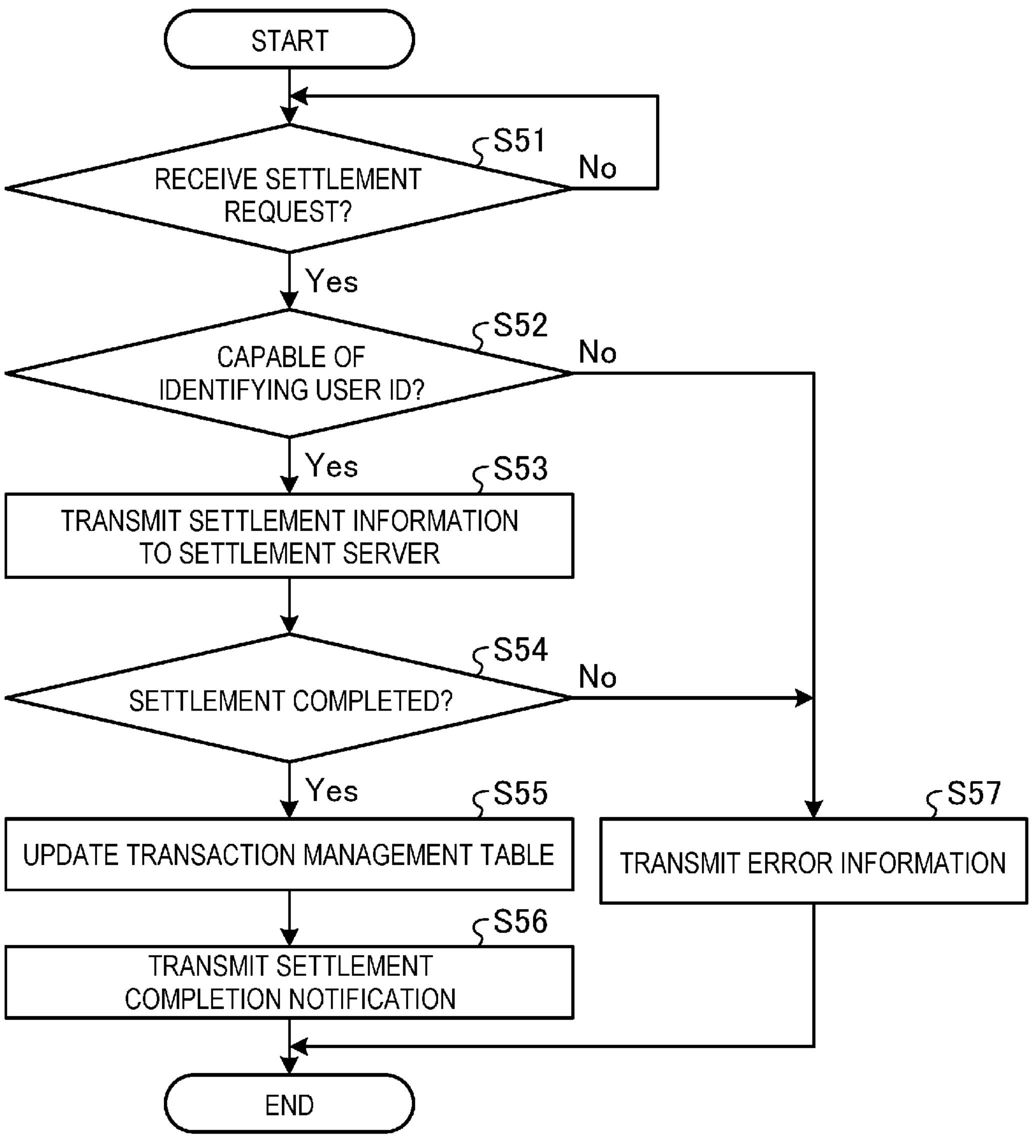

START
S41
RECEIVE SETTLEMENT INSTRUCTION?
No
Yes
S42
READ MERCHANDISE INFORMATION
S43
DISPLAY MERCHANDISE INFORMATION
S44
CALCULATE TRANSACTION AMOUNT
S45
FACE IMAGE DATA ACQUIRED?
No
Yes
S46
TRANSMIT SETTLEMENT REQUEST
END

FIG. 27


START
RECEIVE SETTLEMENT REQUEST?
S51
No
Yes
CAPABLE OF IDENTIFYING USER ID?
S52
No
Yes
TRANSMIT SETTLEMENT INFORMATION TO SETTLEMENT SERVER
S53
SETTLEMENT COMPLETED?
S54
No
Yes
UPDATE TRANSACTION MANAGEMENT TABLE
S55
TRANSMIT SETTLEMENT COMPLETION NOTIFICATION
S56
TRANSMIT ERROR INFORMATION
S57
END

MOBILE SALES SYSTEM AND SERVER DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-025950, filed Feb. 22, 2022, the entire contents of which are incorporated herein by reference.

FIELD

Embodiments described herein relate to a mobile sales system and a server device for a mobile sales system. The mobile sales system includes one or more mobile sales robots or the like which move to customer designated locations to sale merchandise.

BACKGROUND

In the related art, a system including a robot (shopping assistance device) has been proposed. The robot moves in a store that sells merchandise and assists a customer in shopping. The robot in such a system circulates within the store, receives items to be purchased from the customer, and carries the items to a cash register, thereby supporting the shopping performed by the customer.

As described above, the system in the related art only assists the customer while shopping in the store, and construction of a more useful system is desired for a business operator who sells merchandise.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram illustrating an example of a mobile sales system according to an embodiment.

FIGS. **2** to **4** are a perspective views illustrating aspects of an external configuration of a mobile sales device according to an embodiment.

FIG. **5** is a diagram illustrating an example of a hardware configuration of a mobile sales device according to an embodiment.

FIG. **6** is a diagram illustrating an example of a data configuration of a merchandise master according to an embodiment.

FIG. **7** is a diagram illustrating an example of a data configuration of an accommodated merchandise table.

FIG. **8** is a diagram illustrating an example of a hardware configuration of a user terminal.

FIG. **9** is a diagram illustrating an example of a hardware configuration of a server device.

FIG. **10** is a diagram illustrating an example of a data configuration of a robot management table.

FIG. **11** is a diagram illustrating an example of a data configuration of a merchandise management table.

FIG. **12** is a diagram illustrating an example of a data configuration of a user management table.

FIG. **13** is a diagram illustrating an example of a data configuration of a call management table.

FIG. **14** is a diagram illustrating an example of a data configuration of a transaction management table.

FIG. **15** is a block diagram illustrating functional aspects of devices in a mobile sales system.

FIG. **16** is a diagram illustrating an example of a first display screen displayed on a display unit of a user terminal.

FIG. **17** is a diagram illustrating an example of a second display screen.

FIG. **18** is a diagram illustrating an example of a third display screen.

FIG. **19** is a diagram illustrating an example of a fourth display screen.

FIG. **20** is a diagram illustrating an example of a call position designation screen.

FIG. **21** and FIG. **22** are diagrams illustrating an example of a display screen when a mobile sales device arrives at a call position.

FIG. **23** is a diagram illustrating an example of a display screen when a purchase time limit elapses.

FIG. **24** is a sequence chart illustrating an example of an operation of a mobile sales system.

FIG. **25** is a flowchart of call request reception executed by a control unit of a server device.

FIG. **26** is a flowchart of payment flow executed by a control unit of a mobile sales device.

FIG. **27** is a flowchart of settlement flow executed by a control unit of a server device.

DETAILED DESCRIPTION

According to one embodiment, a mobile sales system includes a terminal device and a mobile sales device. Items to be sold can be mounted on the mobile sales device. The terminal device is used to designate a customer call position for the mobile sales device and is configured to transmit position information indicating the customer call position. The mobile sales device is configured to acquire the position information transmitted by the terminal device, move to a location corresponding to the acquired position information, and execute payment processing for an item removed from the mobile sales device by a customer.

Hereinafter, certain example embodiments will be described with reference to drawings. Further, the present disclosure is not limited to the specific example embodiments.

FIG. **1** is a diagram illustrating an example of a configuration of a mobile sales system according to an embodiment. As illustrated in FIG. **1**, a mobile sales system **1** includes a mobile sales robot **10**, user terminals **20**, and server devices **30**. The mobile sales robot **10**, the user terminals **20**, and the server devices **30** are connected to a network N such as a local area network (LAN).

The mobile sales robot **10** is a self-propelled sales device that transports and sells an item of merchandise. The mobile sales robot **10** is an example of a mobile sales device. For example, the mobile sales robot **10** travels within a predetermined range or area such as a shopping mall, and sells merchandise at various positions within the traveling range/area. The mobile sales robot **10** may perform circulate traveling (circuit travel) involving traveling along a predetermined route and/or call (on-demand) traveling toward a call position as designated by a customer. The number of mobile sales robots **10** is not limited.

Here, an external configuration of the mobile sales robot **10** will be described with reference to FIGS. **2** to **4**. FIGS. **2** to **4** are perspective views illustrating an example of the external configuration of the mobile sales robot **10**.

In FIGS. **2** to **4**, the configuration of the mobile sales robot **10** will be described using an X axis, a Y axis, and a Z axis that are orthogonal to one another. Hereinafter, the left side (a Y-axis negative direction side) of the mobile sales robot **10** is also referred to as a front surface side of the mobile sales robot **10**. The right side (a Y-axis positive direction side) of the mobile sales robot **10** is also referred to as a back surface side of the mobile sales robot **10**.

The mobile sales robot **10** includes a main body unit **11**, a moving unit **12**, and a user interface unit **13**. The main body unit **11** is formed of a box-shaped housing whose front surface and back surface sides are open. An accommodation portion **111** is provided inside the main body unit **11**.

The accommodation portion **111** is a space for accommodating items of merchandise C ("merchandises C") to be sold, such as food, and is an example of a mounting portion. In the accommodation portion **111**, shelves **112** capable of displaying the merchandises C are arranged in multiple stages in the upper-lower direction while extending in the front-rear direction of the mobile sales robot **10**. The merchandises C are placed on the shelves **112** by a store clerk or the like who manages the mobile sales robot **10**. The merchandises C accommodated in the accommodation portion **111** are not limited to food and may be items of another type (category) such as medicine or leisure goods. The accommodation portion **111** may accommodate the merchandises C of a plurality of types.

For example, the shelves **112** may be divided into a plurality of regions in order to classify and place the merchandises C of the same type in the same regions. In this case, the shelves **112** may be divided into a plurality of regions using, for example, a tray.

A shelf label **113** for displaying a name and price of the merchandise C is provided on a front surface side of the shelf **112**. The shelf label **113** may be a so-called electronic shelf label or digital signage that digitally displays information. Each shelf **112** is provided with a weight detection unit **159** (see FIG. **5**) capable of detecting a weight of the merchandise C placed on the shelf **112**. The weight detection unit **159** detects that an item placed on the shelf **112** has been removed from the shelf **112** by detecting a weight change. The weight detection unit **159** may also similarly detect when a previously removed item is returned to the shelf **112**.

The shelf label **113** is provided on the front surface side of the shelf **112** in the present embodiment. The shelf label **113** may be provided on a back surface side of the shelf **112**. In the present embodiment, both the front surface side and the back surface side of the main body unit **11** (the accommodation portion **111**) are open, but in other examples just one side (for example, the front surface side) may be open.

A door formed of a light transmissive member such as glass may be attached to one or both of the front surface side and/or the back surface side of the main body unit **11** (the accommodation portion **111**). The inside of the accommodation portion **111** may be accessible via the door. For example, the door may be attached to the front surface side of the main body unit **11**. The back surface side of the main body unit **11** may be covered with a wall surface. In this case, the wall surface on the back surface side may be formed using a light transmissive member such as glass. Accordingly, it is possible to prevent the merchandise in the accommodation portion **111** from falling out of the accommodation portion **111** during movement of the mobile sales robot **10** or the like.

Further, the door may be lockable by providing an electronic lock or the like. Accordingly, for example, it is possible to prevent the merchandise in the accommodation portion **111** from being illegally taken out during the movement of the mobile sales robot **10** or the like.

A light emitting unit **114** and a distance measurement sensor **115** are provided on a front side of the main body unit **11**. The light emitting unit **114** includes a light emitting element such as a light emitting diode (LED), and notifies an operation state of the mobile sales robot **10** by emitting light under the control of a control unit **154**. For example, the light emitting unit **114** emits light to notify people that the mobile sales robot **10** is moving.

A light emission color of the light emitting unit **114** is not limited to a single color. For example, the light emitting unit **114** may change the light emission color or a light emission pattern according to the operation state of the mobile sales robot **10**. The light emitting unit **114** may be provided not only on the front side of the main body unit **11** but also on a rear side thereof.

The distance measurement sensor **115** is a sensor device that detects an object around the mobile sales robot **10** and that detects a distance to the object. A sensing result of the distance measurement sensor **115** is output to the control unit **154** and used for collision avoidance during movement, detection of a user, and the like. The distance measurement sensor **115** can be implemented by, for example, a sensor device that detects an object or that measures a distance using a captured image or an ultrasonic wave, or a sensor device such as light detection and ranging (LiDAR) that detects an object using laser light.

A position where the distance measurement sensor **115** is provided is not limited to the front of the mobile sales robot **10**. For example, the distance measurement sensor **115** may be provided at another position such as the rear of the mobile sales robot **10**. An imaging unit **134** may be used as the distance measurement sensor **115**.

The moving unit **12** is provided below the main body unit **11** and movably supports the main body unit **11**. Specifically, the moving unit **12** includes non-driving wheels **121**, a driving wheel **122**, and a drive unit **123** that drives the driving wheel **122**.

The non-driving wheel **121** is a small wheel. The non-driving wheel **121** freely changes a direction thereof according to a direction of a force generated by rotation of the driving wheel **122**, thereby changing a traveling direction of the mobile sales robot **10** (the moving unit **12**).

The driving wheel **122** is rotationally driven by the drive unit **123** to move the mobile sales robot **10** (the moving unit **12**) forward or backward, or to rotate.

The user interface unit **13** is provided in an upper portion of the main body unit **11**. The user interface unit **13** includes a first display unit **131**, a second display unit **132**, a third display unit **133**, the imaging unit **134**, a disinfection device **135**, and the like.

The first display unit **131** is a display device provided on the front surface side of the mobile sales robot **10**. The second display unit **132** is a display device provided on the back surface side of the mobile sales robot **10**. The third display unit **133** is a display device provided on the rear side of the mobile sales robot **10**. The first display unit **131**, the second display unit **132**, and the third display unit **133** display various types of information related to sales of the merchandise C or the like under the control of the control unit **154**.

The imaging unit **134** is provided on the rear side of the mobile sales robot **10**. The imaging unit **134** is a digital camera including an imaging element such as a charge coupled device (CCD) or a complementary MOS (CMOS). The imaging unit **134** captures an image of the user who uses the mobile sales robot **10** to acquire a face image of the user and the like.

The disinfection device **135** is a device permitting the user of the mobile sales robot **10** to perform a disinfection. In the present disclosure, disinfection refers to killing, removing, or otherwise rendering harmless pathogenic bacteria, viruses, and other microorganisms present on or in an object. The disinfection can be read as degerming, sterilization, or pasteurization.

A device configuration of the disinfection device **135** is not particularly limited. For example, the disinfection device **135** may be a sprayer that sprays a disinfectant such as alcohol. The disinfection device **135** may be an ultraviolet disinfection device that emits ultraviolet light. The disinfection device **135** performs a disinfection operation such as spraying the disinfectant or emitting the ultraviolet light according to a user operation.

An operation state of the disinfection device **135** may be electrically output to the control unit **154**. In this case, for example, the user interface unit **13** or the disinfection device **135** includes a sensor device (also referred to as a disinfection operation detection unit) that detects the disinfection operation of the disinfection device **135**. When the disinfection operation detection unit detects that the disinfection device **135** performs the disinfection operation, the disinfection operation detection unit outputs a detection signal to the control unit **154**.

The configuration of the user interface unit **13** is not limited to the illustrated example. For example, the number and arrangement positions of display devices provided in the user interface unit **13** are not limited to the illustrated example. The user interface unit **13** need not include the disinfection device **135**. The user interface unit **13** may include a reader device that reads information stored in a code symbol such as a two-dimensional code, a reader device that reads information from a card medium such as an IC card or a credit card, or the like.

Referring back to FIG. **1**, the user terminal **20** is a device used by a user of the mobile sales system **1** and is an example of a terminal device. The user terminal **20** is implemented as a portable terminal device such as a smartphone or a tablet terminal. The user terminal **20** receives, from the user, a call request for requesting to call the mobile sales robot **10**, and transmits the received call request to the server device **30**. The number of user terminals **20** is not limited to the example in FIG. **1**.

The server device **30** manages the mobile sales system **1**. The server device **30** is implemented by, for example, an information processing device such as a personal computer (PC). The server device **30** tracks items sold by each mobile sales robot **10**. The server device **30** manages the mobile sales robot **10** and causes the mobile sales robot **10** to travel to the call position designated by a call request received from a user terminal **20**. For example, the server device **30** tracks current positions of the mobile sales robots **10** and the user terminals **20**. When receiving the call request for a mobile sales robot **10** from a user terminal **20**, the server device **30** causes the mobile sales robot **10** designated by the call request to travel toward the position of the user terminal **20**.

A hardware configuration and functional aspects of each device provided in the mobile sales system **1** will be described.

First, a hardware configuration of the mobile sales robot **10** will be described. FIG. **5** is a diagram illustrating an example of a hardware configuration of the mobile sales robot **10**. As illustrated in FIG. **5**, the mobile sales robot **10** includes a central processing unit (CPU) **151**, a read only memory (ROM) **152**, and a random access memory (RAM) **153**.

The CPU **151** is an example of a processor, and integrally controls each unit of the mobile sales robot **10**. The ROM **152** stores various programs. The RAM **153** is a work space in which programs and various types of data are loaded.

The CPU **151**, the ROM **152**, and the RAM **153** are connected via a bus or the like, and constitute the control unit **154**. In the control unit **154**, the CPU **151** operates according to a program stored in a storage unit **162** and loaded in the RAM **153**, thereby executing various processes.

The mobile sales robot **10** includes the light emitting unit **114**, the drive unit **123**, and the imaging unit **134** described above. The mobile sales robot **10** includes a display unit **155**, an operation unit **156**, a sound collection unit **157**, a sensor unit **158**, the weight detection unit **159**, a positioning unit **160**, a communication unit **161**, the storage unit **162**, and the like.

The display unit **155** is a display device such as the first display unit **131**, the second display unit **132**, and the third display unit **133** described above. The display unit **155** can be a liquid crystal display (LCD) or the like. The display unit **155** displays various types of information under the control of the CPU **151**. When the shelf label **113** is an electronic shelf label, the display unit **155** includes the electronic shelf label.

The operation unit **156** is an input device such as a keyboard or a pointing device. The operation unit **156** outputs an operation content received from the user to the CPU **151**. The operation unit **156** may be a touch panel provided on the display screen of the display unit **155**.

The sound collection unit **157** collects a sound around the mobile sales robot **10** and outputs a sound signal of the collected sound to the CPU **151**. The sound collection unit **157** is implemented by, for example, a sound collection device such as a microphone. The sound collection unit **157** collects, for example, a sound for stopping the traveling of the mobile sales robot **10** from a user who wishes to purchase the merchandise during the circulate traveling of the mobile sales robot **10**.

The sensor unit **158** is a sensor device such as the distance measurement sensor **115** and the disinfection operation detection unit described above. The sensor unit **158** outputs a detection result obtained by sensing to the CPU **151**.

The weight detection unit **159** is a weight sensor that detects a weight of the merchandise accommodated in the accommodation portion **111**. Specifically, the weight detection unit **159** is provided in each of the shelves **112**, and detects the weight or a weight change of the merchandise placed on the shelf **112**.

Here, each of the shelves **112** and the weight detection unit **159** provided in the shelf **112** are associated with each other in advance. When the weight change is detected by the weight detection unit **159**, the shelf **112** having the weight change can be identified.

When the shelf **112** is divided into a plurality of regions, the weight detection unit **159** may be provided for each of the divided regions. In this case, each of the divided regions and the weight detection unit **159** provided in the region are associated with each other.

The positioning unit **160** measures a position where the mobile sales robot **10** is present. The positioning unit **160** can be implemented by a positioning device using a positioning technique such as a global positioning system (GPS).

The positioning unit **160** may be implemented by a position measurement device using a positioning technique such as beacon positioning or radio frequency identifier (RFID) positioning. In this case, by providing a terminal corresponding to the positioning technique such as the beacon positioning or the RFID positioning at each position in a range in which the mobile sales robot **10** is movable, the position where the mobile sales robot **10** is present can be measured (identified) by a positioning system formed by the terminal and the positioning unit **160**.

The communication unit **161** is a wireless communication interface connectable to the network N. The communication unit **161** communicates with an external device such as the server device **30** via the network N.

The storage unit **162** includes a storage medium such as a hard disk drive (HDD) or a flash memory, and maintains a stored content even when a power supply is cut off. The storage unit **162** stores programs that can be executed by the CPU **151** and various types of setting information.

The storage unit **162** stores map information **1621**, a merchandise master **1622**, and an accommodated merchandise table **1623**. Here, the map information **1621** is information indicating a map having the range in which the mobile sales robot **10** moves.

The merchandise master **1622** is a data table storing information related to a merchandise to be sold. FIG. **6** is a diagram illustrating an example of a data configuration of the merchandise master **1622**. As illustrated in FIG. **6**, the merchandise master **1622** stores, in association with a merchandise code for identifying a merchandise, merchandise information on the merchandise corresponding to the merchandise code. The merchandise information includes a merchandise name, a type, a price, a weight, image data, and the like.

The data configuration of the merchandise master **1622** is not limited to the example in FIG. **6**. For example, the merchandise master **1622** may store image data representing a feature of the merchandise in the merchandise information.

The storage unit **162** may not store the merchandise master **1622**. In this case, the mobile sales robot **10** refers to a merchandise master **3162** stored in the server device **30**. Therefore, the mobile sales robot **10** can handle the merchandise master **3162** similarly to the configuration of storing the merchandise master **1622**.

The accommodated merchandise table **1623** is a data table for storing information related to items accommodated in the accommodation portion **111**. FIG. **7** is a diagram illustrating an example of a data configuration of the accommodated merchandise table **1623**. As illustrated in FIG. **7**, the accommodated merchandise table **1623** stores, in association with a shelf ID capable of identifying each of the shelves **112** provided in the accommodation portion **111**, a merchandise code of the merchandise placed on the shelf **112** having the shelf ID and the quantity of the placed items.

The data configuration of the accommodated merchandise table **1623** is not limited to the example in FIG. **7**. For example, when the shelf **112** is divided into a plurality of regions, the accommodated merchandise table **1623** may assign an identifier to each of the divided regions, and store the merchandise code of the merchandise placed in the region and the quantity in association with each other.

Next, a hardware configuration of the user terminal **20** will be described. FIG. **8** is a diagram illustrating an example of the hardware configuration of the user terminal **20**. As illustrated in FIG. **8**, the user terminal **20** includes a CPU **211**, a ROM **212**, and a RAM **213**.

The CPU **211** is an example of a processor, and integrally controls each unit of the user terminal **20**. The ROM **212** stores various programs. The RAM **213** is a work space in which programs and various types of data are loaded.

The CPU **211**, the ROM **212**, and the RAM **213** are connected via a bus or the like, and constitute a control unit **214**. In the control unit **214**, the CPU **211** operates according to a control program stored in the storage unit **220** and loaded in the RAM **213**, thereby executing various processes.

The user terminal **20** includes a display unit **215**, an operation unit **216**, an imaging unit **217**, a positioning unit **218**, a communication unit **219**, a storage unit **220**, and the like.

The display unit **215** is a display device such as an LCD screen or the like. The display unit **215** displays various types of information under the control of the CPU **211**. The operation unit **216** is an input device such as a keyboard or a pointing device. The operation unit **216** outputs an operation content received from the user to the CPU **211**. The operation unit **216** may be a touch panel provided on a display screen of the display unit **215**.

The imaging unit **217** can be a digital camera including an imaging element such as a CCD or a CMOS. The imaging unit **217** captures an image of the user of the user terminal **20** to acquire a face image of the user.

The positioning unit **218** measures a position where the user terminal **20** is present. The positioning unit **218** can be implemented by, for example, a positioning device using a positioning technique such as GPS. The positioning unit **218** may be implemented by a position measurement device using a positioning technique such as beacon positioning or RFID positioning, similarly to the above-described positioning unit **160**.

The communication unit **219** is a wireless communication interface connectable to the network N. The communication unit **219** communicates with an external device such as the server device **30** via the network N.

The storage unit **220** includes a storage medium such as an HDD or a flash memory, and maintains a stored content even when the power supply is cut off. The storage unit **220** stores programs that can be executed by the CPU **211** (including application programs related to processes of the mobile sales system **1**) and various types of setting information.

The storage unit **220** stores map information **2201** and the like. Here, similarly to the map information **1621**, the map information **2201** is information indicating a map having the range in which the mobile sales robot **10** moves.

Next, a hardware configuration of the server device **30** will be described. FIG. **9** is a diagram illustrating an example of the hardware configuration of the server device **30**. As illustrated in FIG. **9**, the server device **30** includes a CPU **311**, a ROM **312**, and a RAM **313**.

The CPU **311** is an example of a processor, and integrally controls units of the server device **30**. The ROM **312** stores various programs. The RAM **313** is a work space in which programs and various types of data are loaded.

The CPU **311**, the ROM **312**, and the RAM **313** are connected via a bus or the like, and constitute a control unit **314**. In the control unit **314**, the CPU **311** operates according to a control program stored in the storage unit **316** and loaded in the RAM **313**, thereby executing various processes.

The server device **30** includes a communication unit **315**, a storage unit **316**, and the like. The communication unit **315** is a wired or wireless communication interface connectable to the network N. The communication unit **315** communicates with external devices such as the mobile sales robot **10** and the user terminal **20** via the network N.

The storage unit **316** includes a storage medium such as an HDD or a flash memory, and maintains a stored content even when the power supply is cut off. The storage unit **316** stores programs that can be executed by the CPU **311** and various types of setting information.

The storage unit **316** stores map information **3161** and the merchandise master **3162**. The map information **3161** is information indicating a map having the range in which the mobile sales robot **10** moves. The merchandise master **3162** is a data table storing information related to a merchandise to be sold. The data configuration of the merchandise master **3162** is similar to that of the merchandise master **1622** described above.

The storage unit **316** stores a robot management table **3163**, a merchandise management table **3164**, a user management table **3165**, a call management table **3166**, a transaction management table **3167**, and the like.

The robot management table **3163** is a data table for managing a position and a state of the mobile sales robot **10**. FIG. **10** is a diagram illustrating an example of a data configuration of the robot management table **3163**. As illustrated in FIG. **10**, the robot management table **3163** stores, in association with a robot ID, position information indicating a current position of the mobile sales robot **10** corresponding to the robot ID and state information indicating the state of the mobile sales robot **10**. The robot ID is used for identifying the mobile sales robot **10**, and is an example of mobile sales device identifying information.

Here, the position information may be indicated by coordinate values such as longitude and latitude. The position information may be indicated by a block number, grid number, or the like obtained by dividing a movable range of the mobile sales robot **10** into a plurality of blocks.

Examples of the possible states of the mobile sales robot **10** include a state of "moving" in which the mobile sales robot **10** moves to a call position designated by the user, a state of "waiting for transaction" in which the mobile sales robot **10** arrives at the call position and waits for an operation of starting a transaction, and a state of "transaction" in which a transaction has been started by user operation. Other examples of the possible state of the mobile sales robot **10** include a state of "waiting for call" (capable of responding to a call from a user), such as a time when the mobile sales robot **10** circulates, and a state of "unsellable" in which the mobile sales robot **10** cannot sell merchandise due to the need for merchandise replenishment or the like.

The information stored in the robot management table **3163** is updated according to the movement of the mobile sales robot **10** and the state of the mobile sales robot **10** under the control of an information management unit **3142**.

The merchandise management table **3164** is a data table for managing items sold by each mobile sales robot **10**. FIG. **11** is a diagram illustrating an example of a data configuration of the merchandise management table **3164**. As illustrated in FIG. **11**, the merchandise management table **3164** stores, in association with the robot ID, the merchandise code of the merchandise sold by the mobile sales robot **10** having the robot ID and a stock quantity of the items.

The information stored in the merchandise management table **3164** is updated according to the stock quantity of the items stored by the mobile sales robot **10** under the control of the information management unit **3142**.

The user management table **3165** is a data table for managing users who use the mobile sales system **1**. FIG. **12** is a diagram illustrating an example of a data configuration of the user management table **3165**. As illustrated in FIG. **12**, the user management table **3165** stores, in association with a user ID capable of identifying each user, user information related to the user having the user ID.

The user information includes feature information indicating features of a face of the user and settlement information used for electronic settlement. The feature information is, for example, a face image obtained by capturing an image of the face of the user, or feature data indicating the features of the face, and is used as collation data in a face authentication. The settlement information is information such as an account for using an electronic settlement service contracted in advance by the user. Here, the electronic settlement service includes code settlement, electronic money settlement, credit settlement, and the like. When the electronic settlement service is the credit settlement, the settlement information may be a number of a credit card owned by the user.

The user information is acquired from the user according to a standard member registration method and is registered in the user management table **3165**. The user ID may be automatically assigned, or may be unique information input by the user. The user information may include other information. For example, the user information may include personal information such as a name, an age, and an address of the user. The user information may include a terminal ID of a terminal (the user terminal **20**) used by the user, a communication address of the terminal, and the like.

The call management table **3166** is a data table for managing calls of the mobile sales robot **10** by the user terminal **20**. FIG. **13** is a diagram illustrating an example of a data configuration of the call management table **3166**. As illustrated in FIG. **13**, the call management table **3166** stores a terminal ID of the user terminal **20** that makes the call, a date and time (a call date and time) when the call is received, position information (also simply referred to as a "call position") indicating the call position, authentication information used for authentication for the user who called, a robot ID of the mobile sales robot **10** corresponding to the call, a date and time (an arrival date and time) when the mobile sales robot **10** arrived at the call position, and the like in association with one another.

As the authentication information, for example, a pin code of a predetermined number of digits, a face image of a user, and/or a user ID can be used. In the former case, the pin code may be issued by the control unit **314** or may be designated by the user. In the latter case, the face image or the user ID transmitted from the user terminal **20** of the user who called, the user ID identified by the control unit **314** from the user management table **3165** based on the transmitted face image, or the like can be used. In the present embodiment, a configuration in which the control unit **314** issues the pin code when receiving the call will be described.

The transaction management table **3167** is a data table for managing transactions via the mobile sales robot **10**. FIG. **14** is a diagram illustrating an example of a data configuration of the transaction management table **3167**. As illustrated in FIG. **14**, the transaction management table **3167** stores, in association with a transaction ID for identifying a transaction, a robot ID of the mobile sales robot **10** with which the transaction is performed, a user ID of the user who performs the transaction, a merchandise code of the merchandise purchased by the user, a settlement flag for determining whether the settlement is completed, and the like.

Next, a functional configuration of each device constituting the mobile sales system **1** will be described. FIG. **15** is a block diagram illustrating the functional configuration of devices constituting the mobile sales system **1**. First, the user terminal **20** will be described.

The control unit **214** of the user terminal **20** functions as a transmission and reception unit **2141**, an input reception unit **2142**, a self-position identifying unit **2143**, and a display control unit **2144** when the CPU **211** operates according to a control program stored in the ROM **212** or the storage unit **220**. The functions may be implemented by hardware such as a dedicated circuit.

The transmission and reception unit **2141** transmits and receives various types of information to and from an external device such as the server device **30** via the communication unit **219**. A function and operation of the transmission and reception unit **2141** to "receive" information from the server device **30** or the like can also be rephrased as "acquiring" information.

For example, the transmission and reception unit **2141** receives, from the server device **30**, each piece of information related to the robot ID, the current position, and the state of the mobile sales robot **10**. The transmission and reception unit **2141** transmits a merchandise information request or a call request to the server device **30**. The merchandise information request designates the mobile sales robot **10** and requests information on the merchandise accommodated in the mobile sales robot **10**, that is, the merchandise to be sold. The merchandise information request includes information such as a terminal ID of the user terminal **20** which is a request source, and a robot ID of the mobile sales robot **10** to be designated.

The call request designates the mobile sales robot **10** and requests a call of the mobile sales robot **10**. The call request includes information such as the terminal ID of the user terminal **20** which is the request source, the robot ID of the mobile sales robot **10** to be designated, and the call position. In a configuration in which the user designates authentication information, the call request further includes the designated authentication information. In the present embodiment, the call position is a position where the user terminal **20** is present, that is, a position measured by the positioning unit **218**. The call position desired by the user may be designated by the user terminal **20**.

The transmission and reception unit **2141** receives the merchandise information transmitted by the server device **30** according to the merchandise information request. Furthermore, the transmission and reception unit **2141** receives a pin code issued by the server device **30** in association with the call request. The pin code is for authenticating the user who called the mobile sales robot **10** is the user who now intends to start shopping at the mobile sales robot **10**.

The transmission and reception unit **2141** receives an arrival notification or a settlement completion notification from the server device **30**. The arrival notification indicates that the mobile sales robot **10** arrives at the call position designated by the call request. The settlement completion notification indicates that the settlement related to the purchase of the merchandise accommodated in the mobile sales robot **10** is completed.

The input reception unit **2142** receives input information based on an operation of the operation unit **216**. For example, the input reception unit **2142** receives the merchandise information request or the call request from the operation unit **216**.

The self-position identifying unit **2143** identifies the position of the user terminal **20** in an area represented by the map information **2201** based on an output of the positioning unit **218**.

The display control unit **2144** causes the display unit **215** to display various types of information. Specifically, the display control unit **2144** causes the display unit **215** to display various screens (a graphic user interface (GUI)) for supporting purchase of the merchandise sold by the mobile sales robot **10**. For example, the display control unit **2144** causes the display unit **215** to display a display screen for supporting the call of the mobile sales robot **10**.

An example of a display screen displayed by the display control unit **2144** will be described with reference to FIGS. **16** and **17**. FIG. **16** is a diagram illustrating an example of a first display screen displayed on the display unit **215** of the user terminal **20**. FIG. **17** is a diagram illustrating an example of a second display screen displayed on the display unit **215** of the user terminal **20**.

The display control unit **2144** displays the mobile sales robots **10** in a selectable manner on the map in response to positions where the mobile sales robots **10** are present, and then displays the merchandise information of the items mounted on the selected mobile sales robots **10** together with a call button. Specifically, when a merchandise information request menu is selected from a menu selection screen or the like, the display control unit **2144** displays the first display screen illustrated in FIG. **16** on the display unit **215**.

The first display screen includes a map display region **2151**. The display control unit **2144** displays, based on the map information **2201**, a map having the range in which the mobile sales robot **10** is movable in the map display region **2151**. Here, the display control unit **2144** may display a map of the vicinity of the position where the user terminal **20** is present based on the position information identified by the self-position identifying unit **2143**, or may display a map of the vicinity of the position where the mobile sales robot **10** is present based on the position information of the mobile sales robot **10** provided from the server device **30**. The display control unit **2144** changes the range of the map to be displayed in the map display region **2151** according to an operation (for example, a scroll operation, and an enlargement and reduction operation) on the map.

The display control unit **2144** displays robot marks **2152** representing the mobile sales robots **10** in a manner of being superimposed on the map of the map display region **2151**. Specifically, the display control unit **2144** displays the robot marks **2152** at positions where the mobile sales robots **10** are present on the map. Robot identification information such as the robot ID is also displayed on the robot mark **2152**.

The user can select the corresponding mobile sales robot **10** by touching any one of the robot marks **2152** displayed on the first display screen. When the input reception unit **2142** receives the designation of the mobile sales robot **10**, the transmission and reception unit **2141** transmits the merchandise information request including the terminal ID of the user terminal **20** and the robot ID of the designated mobile sales robot **10** to the server device **30**. The server device **30** that receives the merchandise information request reads the merchandise information on the merchandise sold by the mobile sales robot **10** having the designated robot ID from the merchandise management table **3164** and transmits the merchandise information to the user terminal **20**.

When the transmission and reception unit **2141** receives the merchandise information, the display control unit **2144** displays the merchandise information in association with the robot mark **2152** of the selected mobile sales robot **10**, as illustrated in the second display screen in FIG. **17**. That is, the second display screen is a screen to be displayed when any one of the robot marks **2152** displayed on the first display screen is touched.

As illustrated in FIG. **17**, the second display screen includes the map display region **2151** and a merchandise information display region **2153**. Similarly to the map display region **2151** on the first display screen, the map display region **2151** is a region for displaying a map based on the map information **2201**. The map display region **2151** is smaller than the map display region **2151** on the first display screen. The display control unit **2144** causes the robot mark **2152** touched on the first display screen and a map around the robot mark **2152** to be displayed in the map display region **2151**.

The merchandise information display region **2153** is a region for displaying the merchandise information on the merchandise mounted on the selected mobile sales robot **10**. The display control unit **2144** causes the merchandise information transmitted from the server device **30** according to the selection of the mobile sales robot **10** to be displayed in the merchandise information display region **2153**. Specifically, the display control unit **2144** causes merchandise information to be displayed side by side in an upper-lower direction on a per merchandise item basis. The displayed merchandise information includes a merchandise image, a merchandise name, a price, and the like. The merchandise information display region **2153** can be scrolled in the upper-lower direction. The display control unit **2144** updates the merchandise information to be displayed according to a scroll operation. Accordingly, the merchandise information display region **2153** can display the merchandise information of all the items mounted on the mobile sales robot **10**.

The second display screen includes a call button **2154** for calling the selected mobile sales robot **10**. The user can check, on the second display screen, the items that is mounted on the mobile sales robot **10** selected on the first display screen, and can call the mobile sales robot **10** by touching the call button **2154** when there is a merchandise that the user desires to purchase. By calling the mobile sales robot **10** using the first display screen and the second display screen, the user can select the mobile sales robot **10** to be called while checking the position of the mobile sales robot **10**. Therefore, for example, the mobile sales robot **10** can be efficiently called, such as calling the mobile sales robot **10** closest to the position of the user.

Specifically, when the input reception unit **2142** receives the operation of the call button **2154** on the second display screen, the transmission and reception unit **2141** transmits, to the server device **30**, the call request including the terminal ID of the user terminal **20**, the robot ID of the designated mobile sales robot **10**, the position information (the call position) identified by the self-position identifying unit **2143**, and the like. The server device **30** that receives the call request causes the designated mobile sales robot **10** (as corresponding to the designated robot ID) to travel to the designated call position. The server device **30** predicts, based on the current position and the call position of the designated robot ID, the number of call requests for the robot ID registered in the call management table **3166**, and the like, a time until the mobile sales robot **10** corresponding to the robot ID arrives at the call position (also referred to as an arrival time), and transmits the predicted arrival time to the user terminal **20** which is the request source.

The second display screen includes an arrival time display region **2155** for displaying a time until the selected mobile sales robot **10** arrives at the position of the user terminal **20**. The display control unit **2144** displays the arrival time transmitted from the server device **30** according to the selection of the mobile sales robot **10** in the arrival time display region **2155**. In the arrival time display region **2155**, the robot ID or the like of the mobile sales robot **10** scheduled to arrive may be displayed.

A timing of displaying the arrival time is not limited to a timing after the operation of the call button **2154**. For example, the display control unit **2144** may display the arrival time at a timing when the mobile sales robot **10** is selected on the first display screen. In this case, the transmission and reception unit **2141** may include the position information identified by the self-position identifying unit **2143** in the merchandise information request and transmit the merchandise information request to the server device **30**, thereby acquiring the arrival time from the server device **30** at the timing when the mobile sales robot **10** is selected.

The second display screen displays the mobile sales robot **10** selected on the first display screen in an identifiable manner. Specifically, in the map display region **2151**, the robot mark **2152** to which the display of "robot B" is added is displayed together with the map around the robot mark **2152**. The "robot B" is the information for identifying the mobile sales robot **10** selected on the first display screen. Accordingly, the user can check the position, the identification information, the merchandise information on the merchandise to be mounted, and the like of the selected mobile sales robot **10** on one screen. The information for identifying the mobile sales robot **10** may be a unique number, name, or the like set for each mobile sales robot **10**.

A method for displaying the information for identifying the selected mobile sales robot **10** is not limited to the above. For example, the information for identifying the mobile sales robot **10** in association with the arrival time may be displayed in the arrival time display region **2155**. The information for identifying the mobile sales robot **10** may be displayed in the merchandise information display region **2153**.

The display screen related to the call of the mobile sales robot **10** is not limited to the examples in FIGS. **16** and **17**. Additional display screens will be described with reference to FIGS. **18** and **19**.

FIG. **18** is a diagram illustrating an example of a third display screen displayed on the display unit **215** of the user terminal **20**. On the third display screen, the selection of the mobile sales robot **10**, the display of the merchandise information on the merchandise mounted on the mobile sales robot **10**, and the call of the mobile sales robot **10** can be performed on the same screen.

Specifically, the third display screen includes a robot selection region **2156**, the merchandise information display region **2153**, the call button **2154**, and the arrival time display region **2155**.

The robot selection region **2156** is a region for displaying an image (a robot mark) corresponding to each mobile sales robot **10** in a selectable manner. The merchandise information display region **2153** is a region for displaying the merchandise information on the merchandise sold by the mobile sales robot **10** having the robot mark selected in the robot selection region **2156**.

Here, images of the robot mark displayed in the robot selection region **2156** corresponds to the mobile sales robots **10** traveling in a predetermined region. The display control unit **2144** causes the robot mark corresponding to each robot ID to be displayed in the robot selection region **2156** based on the robot ID, the position information corresponding to the robot ID, and the like transmitted from the server device **30**. The robot selection region **2156** can be scrolled in a left-right direction. The display control unit **2144** updates the robot mark to be displayed according to the scroll operation. Accordingly, the robot selection region **2156** can display all the mobile sales robots **10**.

The display control unit **2144** may display the robot ID and the state of the corresponding mobile sales robot in association with each robot mark.

The user can select the mobile sales robot **10** by touching any one of the robot marks displayed in the robot selection region **2156**. According to the selection of the mobile sales robot **10**, the transmission and reception unit **2141** transmits the merchandise information request designating the robot ID of the selected mobile sales robot **10** to the server device **30**, and acquires the merchandise information from the server device **30**. Then, the display control unit **2144** displays the merchandise information on the merchandise sold by the mobile sales robot **10** selected in the robot selection region **2156** in the merchandise information display region **2153**.

Accordingly, the user can browse the merchandise information on the merchandise accommodated in the selected mobile sales robot **10** through the merchandise information display region **2153**. The user can check the merchandise mounted on the mobile sales robot **10** selected in the robot selection region **2156** in the merchandise information display region **2153**, and can call the mobile sales robot **10** by touching the call button **2154** when there is a merchandise that the user desires to purchase. Therefore, the user can select the mobile sales robot **10**, browse the merchandise information on the merchandise mounted on the selected mobile sales robot **10**, and call the mobile sales robot **10** without switching the screen.

In the arrival time display region **2155**, an arrival time of the mobile sales robot **10** to be called (or selected) is displayed similarly to the second display screen.

FIG. **19** is a diagram illustrating an example of a fourth display screen displayed on the display unit **215** of the user terminal **20**. On the fourth display screen, instead of explicitly selecting the mobile sales robot **10**, by selecting a merchandise or a merchandise type, it is possible to select the mobile sales robot **10** on which the selected merchandise or type is mounted. In the present embodiment, it is possible to select the merchandise type.

Specifically, the fourth display screen includes a merchandise type selection region **2157**, the merchandise information display region **2153**, the call button **2154**, and the arrival time display region **2155**.

The merchandise type selection region **2157** is a region for displaying icons indicating merchandise types mounted on the mobile sales robot **10**. The display control unit **2144** displays icons indicating the merchandise types mounted on the mobile sales robot **10** in the merchandise type selection region **2157** in a selectable manner. Here, the merchandise types to be displayed as the icons may be determined in advance. The merchandise types to be displayed as the icons may be determined based on a merchandise type) included in the merchandise information for items sold by each of the mobile sale robots **10** from the server device **30**.

When the user touches one of the icons displayed in the merchandise type selection region **2157**, the transmission and reception unit **2141** transmits identification information for identifying the merchandise type corresponding to the icon to the server device **30**. The server device **30** refers to the merchandise management table **3164** and extracts the mobile sales robot **10** on which the merchandise of the appropriate type is mounted. In the present example, it is assumed that the server device **30** includes a management table in which merchandise type and merchandise are associated with each other. The mobile sales robot **10** selected by the server device **30** may be a mobile sales robot **10** on which only merchandise of the merchandise type is mounted or a mobile sales robot **10** on which merchandise of multiple types including the selected type is mounted.

Then, the server device **30** selects one of the extracted mobile sales robots **10** (for example, the mobile sales robot **10** located closest to the position of the user terminal **20**). The server device **30** transmits the merchandise information for the merchandise mounted on the selected mobile sales robot **10** to the user terminal **20**.

Accordingly, the user can select the mobile sales robot **10** on which the merchandise of the merchandise type indicated by the icon is mounted by touching one of the icons displayed in the merchandise type selection region **2157**. The user can browse the merchandise information on the merchandise accommodated in the selected mobile sales robot **10** through the merchandise information display region **2153**. The merchandise information display region **2153** is similar to the merchandise information display region **2153** on the second display screen illustrated in FIG. **17**, and thus the repeated description will be omitted.

Any of the method for calling the mobile sales robot **10** based on the first display screen and the second display screen, the method for calling the mobile sales robot **10** based on the third display screen, and the method for calling the mobile sales robot **10** based on the fourth display screen may be used. For example, a configuration may be adopted in which the method of calling can be switched according to an operation of a switching button or the like.

The display screens in FIGS. **16** to **19** may be used in combination. For example, the display control unit **2144** first displays the fourth display screen. When the merchandise type is selected on the fourth display screen, the display control unit **2144** displays the first display screen on which the mobile sales robot **10** is selectable. The merchandise of the selected merchandise type is mounted on the mobile sales robot **10**. Then, the display control unit **2144** may display the second display screen based on the mobile sales robot **10** selected on the first display screen. In this case, the fourth display screen to be displayed first may include only the merchandise type selection region **2157**.

As described above, the user terminal **20** transmits the merchandise information request, receives and displays the merchandise information, and transmits the call request according to a user operation. At this time, the call position of the mobile sales robot **10** is the position of the user. The display control unit **2144** may display a call position designation screen on which the call position of the mobile sales robot **10** can be designated before the call request is transmitted to the server device **30**.

FIG. **20** is a diagram illustrating an example of the call position designation screen displayed on the display unit **215** of the user terminal **20**. The call position designation screen includes, for example, the map display region **2151** similar to that of the first display screen. The display control unit **2144** displays the robot mark **2152** of the selected mobile sales robot **10** on the map of the map display region **2151**. On the call position designation screen, the user can designate any position on the map as the call position of the mobile sales robot **10**.

The display control unit **2144** displays a designated position mark **2158** at the call position on the map designated by the user. In the designated position mark **2158**, the current position of the user, that is, the current position of the user terminal **20** is set as a default. The user can move the designated position mark **2158** to a desired position by, for example, a drag-and-drop operation. Thereafter, when the user performs a predetermined operation, the transmission and reception unit **2141** transmits a call request including the designated call position to the server device **30**. For example, when the user touches a call confirmation button or the like displayed on the map display region **2151**, the user terminal **20** may transmit the call request to the server device **30**.

After the transmission and reception unit **2141** transmits the call request, the display control unit **2144** displays the current position of the mobile sales robot **10** on the map of the map display region **2151** using the robot mark **2152** or the like. The position information is provided from the server device **30**.

Accordingly, the user can easily check the current position of the mobile sales robot **10** selected by viewing the map of the map display region **2151**.

When the selected mobile sales robot **10** moves to the call position, the display control unit **2144** causes the display unit **215** to display a display screen for notifying that the mobile sales robot **10** arrived at the call position.

The display screen displayed on the display unit **215** by the display control unit **2144** after the mobile sales robot **10** arrives at the call position will be described with reference to FIGS. **21** to **23**.

FIG. **21** is a diagram illustrating an example of the display screen displayed on the display unit **215** when the mobile sales robot **10** arrives at the call position. When the mobile sales robot **10** arrives at the call position, a push notification is given from the server device **30** to the user terminal **20**. According to the notification from the server device **30**, the display control unit **2144** causes the display unit **215** to pop-up display a message A notifying that the mobile sales robot **10** arrives. FIG. **21** illustrates an example in which the message A is pop-up displayed on the above-described second display screen (the same applies to FIGS. **22** and **23**).

As described above, when the mobile sales robot **10** arrives at the call position, the display control unit **2144** performs a notification using the display screen. Accordingly, the user can easily recognize that the mobile sales robot **10** arrives at the call position by viewing the display screen (the message A). The notification of the arrival is not limited to the method using the display screen. For example, the display control unit **2144** may cooperate with a notification unit such as a speaker or a vibrator provided in the user terminal **20** to notify that the mobile sales robot **10** arrives at the call position by sound or vibration. The display control unit **2144** may display, on a pop-up screen, a robot number and the robot ID displayed on the mobile sales robot **10** that arrives. Accordingly, the user can more easily recognize the mobile sales robot **10** called by the user. The method for displaying the robot number and the like on the mobile sales robot **10** is not limited.

In the present embodiment, the arrival at the call position is notified according to the notification from the server device **30**, and is not limited thereto. The user terminal **20** may independently perform the notification by determining whether the mobile sales robot arrives by the user terminal **20**. In this case, for example, the control unit **214** of the user terminal **20** determines whether the position information of the mobile sales robot **10** to be called, which is provided from the server device **30**, falls within a predetermined range with the position information (the call position) of the designated position mark **2158** as a base point. Then, when the mobile sales robot **10** enters the predetermined range, the control unit **214** determines that the mobile sales robot **10** arrives at the call position. In this case, the display control unit **2144** causes the display unit **215** to display a display screen for notifying that the mobile sales robot **10** arrives at the call position according to the determination by the control unit **214**.

The display screen displayed when the mobile sales robot **10** arrives at the call position is not limited to FIG. **21**. Other information may be displayed, for example, as illustrated in FIG. **22**.

FIG. **22** is a diagram illustrating another example of the display screen displayed by the display unit **215** when the mobile sales robot **10** arrives at the call position. The display screen in FIG. **22** is a display example when a time related to the purchase of the merchandise per user is limited. A time limit related to the purchase of the merchandise is also referred to as a "purchase time limit".

For example, when it is assumed that there are a large number of users who desire to use the mobile sales robot **10**, it is desirable to set the purchase time limit for each individual user. This is because there may be users who call the mobile sales robot **10** out of curiosity even though these users have no intention of purchasing the merchandise. The purchase time limit need not always be set, or the setting may be switched between valid and invalid states.

In the present example, when the mobile sales robot **10** arrives at the call position, the display control unit **2144** causes the display unit **215** to pop-up display a message B including a message notifying the user of the arrival of the mobile sales robot **10** at the call position and a message notifying the user of the purchase time limit. At this time, the transmission and reception unit **2141** notifies the server device **30** that the mobile sales robot **10** arrived at the call position. Here, the pop-up displayed purchase time limit may be the purchase time limit itself set in advance, or may be a remaining time of the purchase time limit. The remaining time of the purchase time limit can be derived by the control unit **214** by subtracting a time (an elapsed time) obtained by subtracting a time point at which the mobile sales robot **10** arrives at the call position from a current time point from the purchase time limit set in advance.

Accordingly, the user can recognize, based on the message B, that the mobile sales robot **10** arrives at the call position and that there is a purchase time limit.

In the present example, as illustrated in FIG. **22**, the display control unit **2144** may also display a time extension button C for extending the purchase time limit. The user can extend the purchase time limit by some amount by touching the time extension button C.

For example, when the time extension button C is operated, the display control unit **2144** adds an extension time set in advance to the remaining time of the purchase time limit being displayed, corrects the remaining time of the purchase time limit, and displays the corrected remaining time. Then, the transmission and reception unit **2141** transmits an extension notification to the server device **30**. The server device **30** recognizes a sales completion time at the call position based on the arrival notification and the extension notification received from the mobile sales robot **10** when the mobile sales robot **10** arrives at the call position. Specifically, the server device **30** recognizes, as the sales completion time, a time point at which the purchase time limit and the extension time elapse from a time point at which the arrival notification is received. The server device **30** can instruct the mobile sales robot **10** to move based on the recognized sales time and the like. For example, when the control unit **214** of the mobile sales robot **10** receives the operation of the time extension button C, the control unit **214** may notify the server device **30** of the extension using the state information by setting the state of the mobile sales robot **10** to a state in which the purchase time limit is extended by the user operation (for example, "during transaction extension").

The method for limiting the purchase of the merchandise by the mobile sales robot **10** when the purchase time limit elapses can be freely set. For example, the settlement of the merchandise subjected to the merchandise registration within the purchase time limit may be enabled, and the merchandise taken out after the purchase time limit elapses may not be registered.

Accordingly, it is possible not to impose an unnecessary time limit on the user who actually purchases merchandise.

In the present example, the display control unit **2144** causes the display unit **215** to display a display screen to notify that the purchase time limit elapses without shopping, when, for example, the merchandise is not taken out. FIG. **23** is a display screen displayed when the purchase time limit elapses without shopping by the user. The display screen is displayed when, for example, no merchandise is taken out from the shelf **112** within the purchase time limit. When the purchase time limit elapses without shopping by the user, the display control unit **2144** causes the display unit **215** to display a message D notifying that the sales of the merchandise performed by the mobile sales robot **10** is cancelled.

Accordingly, the user can recognize that the mobile sales robot **10** moves to the next call position or the like.

In the present embodiment, the call request is transmitted from the user terminal **20** carried by the user to the server device **30**, and is not limited thereto. The call request may be transmitted from a terminal device provided in a facility in which the mobile sales robot **10** travels to the server device **30**. In this case, it is desirable that the terminal device allows the user to set the call position to any place.

Referring back to FIG. **15**, the functional aspects of the server device **30** will be described. The control unit **314** of the server device **30** functions as a transmission and reception unit **3141**, an information management unit **3142**, a collation unit **3143**, a settlement unit **3144**, and a robot management unit **3145** when the CPU **311** operates according to a control program stored in the ROM **312** or the storage unit **316**. The functions may be implemented by hardware such as a dedicated circuit.

The transmission and reception unit **3141** transmits and receives various types of information to and from an external device such as the user terminal **20** or each mobile sales robot **10** via the communication unit **315**. For example, the transmission and reception unit **3141** receives, from each mobile sales robot **10**, position information indicating the position of the mobile sales robot **10** and state information indicating the state of the mobile sales robot **10**. The transmission and reception unit **3141** transmits, to the user terminal **20**, the position information indicating the position of each mobile sales robot **10** and state information indicating the state of each mobile sales robot **10**. The transmission and reception unit **3141** receives a merchandise information request from the user terminal **20**, and transmits merchandise information corresponding to the merchandise information request to the user terminal **20**.

The transmission and reception unit **3141** receives a call request from the user terminal **20** and transmits a movement instruction to the mobile sales robot **10** identified by the robot ID included in the call request. The movement instruction instructs the mobile sales robot **10** to move to the call position included in the call request from the user terminal **20**. The movement instruction includes information such as the terminal ID and the call position of the user terminal **20** that transmits the call request.

The transmission and reception unit **3141** receives an arrival notification from the mobile sales robot **10** and transmits the arrival notification to the user terminal **20** which is a request source of the call request of the mobile sales robot **10**. The arrival notification indicates that the mobile sales robot **10** arrives at the designated call position. The arrival notification includes information such as the robot ID of the mobile sales robot that arrives at the call position.

The transmission and reception unit **3141** transmits a pin code (or other authentication information) to the user terminal **20** in response to the call request, and receives a pin code input to the mobile sales robot **10** by the user from the mobile sales robot **10**. When the collation unit **3143** collates the pin code transmitted to the user terminal **20** with the pin code input to the mobile sales robot **10**, the transmission and reception unit **3141** transmits an authentication notification to the mobile sales robot **10**. The authentication notification indicates that the user who called the mobile sales robot **10** and the user who intends to start shopping at the mobile sales robot **10** match with each other.

The transmission and reception unit **3141** receives a settlement request from the mobile sales robot **10** and transmits a settlement completion notification to the mobile sales robot **10** and the user terminal **20** when the settlement is successfully performed according to the settlement request. The settlement request is a request for price settlement for the merchandise purchased by the user at the mobile sales robot **10**. The settlement request includes information necessary for the settlement executed by the server device **30**. The information necessary for the settlement is face image data of the user in the present embodiment. Instead of the face image data, the user ID or the like may be used. The settlement completion notification indicates that the settlement according to the settlement request is completed. The settlement completion notification includes information such as a transaction ID for identifying the transaction after settlement.

The information management unit **3142** stores, based on the information received by the transmission and reception unit **3141**, information in the storage unit **316** or updates various types of information stored in the storage unit **316**. The information management unit **3142** appropriately reads various types of information transmitted by the transmission and reception unit **3141** from the storage unit **316**.

For example, the information management unit **3142** updates the robot management table **3163** based on the position information and the state information received by the transmission and reception unit **3141** from the mobile sales robot **10**. When the transmission and reception unit **3141** receives the merchandise information request, the information management unit **3142** reads the merchandise code corresponding to the robot ID included in the merchandise information request from the merchandise management table **3164**. Then, the information management unit **3142** reads the merchandise information corresponding to the merchandise code with reference to the merchandise master **3162**.

When the transmission and reception unit **3141** receives the call request or the arrival notification, the information management unit **3142** stores information in the call management table **3166** as appropriate. When the transmission and reception unit **3141** receives the settlement request, the information management unit **3142** stores various types of information in the transaction management table **3167** and reads the settlement information from the user management table **3165**. Specifically, the information management unit **3142** reads the settlement information from the user management table **3165** based on the face image data included in the settlement request. When the settlement has been completed by the settlement unit **3144**, the information management unit **3142** updates the settlement flag of the transaction management table **3167**.

The information management unit **3142** manages a movement order of the mobile sales robot **10** to the call position. Specifically, the information management unit **3142** moves the mobile sales robot **10** to the call position designated by the call request in order from the call request having the earliest call date and time with respect to the call request whose arrival date and time is not registered in the call management table **3166**.

The collation unit **3143** determines whether the user who called the mobile sales robot **10** matches with the user who now intends to start shopping at the mobile sales robot **10**. Specifically, when the transmission and reception unit **3141** receives a call request from the user terminal **20**, the collation unit **3143** issues a pin code associated with the call request. Then, the collation unit **3143** compares whether a pin code received from the mobile sales robot **10** matches with the issued pin code.

The collation unit **3143** may perform a comparison between the user who called the mobile sales robot **10** and the user who now intends to start shopping at the mobile sales robot **10** using face authentication. In this case, the transmission and reception unit **3141** receives the face image data from the mobile sales robot. Then, the collation unit **3143** refers to the user management table **3165**, and performs the comparison based on feature data of the received face image data and feature data of the face of the user who called.

In the mobile sales system **1**, it may not always be essential to confirm that the user who called the mobile sales robot **10** matches with the user who indicates an intention to start shopping at the mobile sales robot **10**. In other words, the server device **30** may omit the function of the collation unit **3143** in some examples.

The settlement unit **3144** executes the settlement related to the settlement request received by the transmission and reception unit **3141** from the mobile sales robot **10**. In the present embodiment, the settlement unit **3144** executes the electronic settlement by the face authentication. Specifically, the settlement unit **3144** performs the settlement with a settlement business operator using the settlement information read by the information management unit **3142** based on the face image data. The settlement unit **3144** executes the settlement by communicating with a settlement server of the settlement business operator. The settlement unit **3144** may execute the electronic settlement by credit card settlement or two-dimensional code settlement.

The robot management unit **3145** sets, based on the call request from the user terminal **20**, the position information and the state information from each mobile sales robot **10**, and the like, a transmission destination to which the transmission and reception unit **3141** transmits the information and a transmission content. For example, when the transmission and reception unit **3141** receives the call request from the user terminal **20**, the robot management unit **3145** sets the mobile sales robot **10** having the robot ID included in the call request as the transmission destination of the movement instruction, and sets the call position included in the call request as the transmission content. For example, the robot management unit **3145** checks the state information of each mobile sales robot **10** stored in the robot management table **3163**, and selects the mobile sales robot **10** corresponding to the call request from among the mobile sales robots **10** in the state of “waiting for call”.

In addition, the robot management unit **3145** sets the user terminal **20**, which is the request source of the call request, as the transmission destination, calculates the arrival time until the selected mobile sales robot **10** arrives at the call position, and sets the arrival time as the transmission content. The arrival time is calculated based on, for example, the number of call requests (still waiting requests whose arrival date and time are not registered) to the corresponding mobile sales robot **10** registered in the call management table **3166**, the purchase time limit in one transaction set in advance, and the position information and the state information of the mobile sales robot **10** registered in the robot management table **3163**.

The robot management unit **3145** refers to the merchandise management table **3164**, sends the mobile sales robot **10** having the robot ID whose stock quantity has become zero for any or all of the merchandise to be mounted a movement instruction, and sets the position at which the merchandise is replenished as a movement destination.

A part or all of the functions of the transmission and reception unit **3141**, the information management unit **3142**, the collation unit **3143**, the settlement unit **3144**, and the robot management unit **3145** described above may be provided in the mobile sales robot **10**. In this case, the user terminal **20** transmits and receives various types of information to and from the mobile sales robot **10**.

Next, the functional configuration of the mobile sales robot **10** will be described. The control unit **154** of the mobile sales robot **10** functions as a transmission and reception unit **1541**, an input reception unit **1542**, a self-position identifying unit **1543**, an information processing unit **1544**, a drive control unit **1545**, and a display control unit **1546** when the CPU **151** operates according to a control program stored in the ROM **152** or the storage unit **162**. In some examples, described functions may be implemented by hardware such as a dedicated circuit.

The transmission and reception unit **1541** transmits and receives various types of information to and from an external device such as the server device **30** via the communication unit **161**. For example, the transmission and reception unit **1541** transmits, to the server device **30**, position information indicating a position of the transmission and reception unit **1541** and state information indicating a state of the transmission and reception unit **1541** as needed.

The transmission and reception unit **1541** transmits an authentication request to the server device **30** and receives an authentication notification from the server device **30**. The authentication request is for requesting authentication of a pin code input to the input reception unit **1542**, and includes the pin code and the robot ID that are received by the input reception unit **1542**. The authentication notification is trigger information for permitting a process related to sales of the merchandise. Therefore, the transmission and reception unit **1541** functions as an input unit to which the trigger information is input. The transmission and reception unit **1541** transmits a settlement request to the server device **30** and receives a settlement completion notification from the server device **30**.

The input reception unit **1542** receives input information based on an operation of the operation unit **156**. For example, the input reception unit **1542** receives a pin code or a settlement instruction input by an operation of the user.

The self-position identifying unit **1543** identifies the position of the mobile sales robot **10** in the map information **1621** based on an output of the positioning unit **160**.

The information processing unit **1544** executes various processes based on various types of information input to the mobile sales robot **10**.

For example, when the transmission and reception unit **1541** receives a movement instruction from the server device **30**, the information processing unit **1544** determines a movement route to the call position based on the map information **1621**.

When the weight detection unit **159** detects the weight change of the shelf **112**, the information processing unit **1544** reads the merchandise code corresponding to the shelf ID of the shelf **112** having the weight change from the accommodated merchandise table **1623**, and identifies the merchandise taken out from the shelf **112** or the merchandise returned to the shelf **112**. The information processing unit **1544** functions as a detection unit that detects the merchandise taken out from the mounting portion.

For example, when the weight change is a decrease in weight, the information processing unit **1544** reads the merchandise code stored in association with the shelf ID of the shelf **112** having the weight change. Then, the information processing unit **1544** refers to the merchandise master **1622** and identifies the merchandise code corresponding to the decreased weight among weights associated with the read merchandise codes, thereby identifying the merchandise taken out from the shelf **112**. In this case, the information processing unit **1544** reads merchandise information corresponding to the identified merchandise code from the merchandise master **1622** and registers the merchandise information in the RAM **153** (also referred to as "merchandise registration"). That is, the information processing unit **1544** also functions as a registration unit that executes the merchandise registration. The information processing unit **1544** subtracts the quantity of the identified merchandise codes stored in the accommodated merchandise table **1623** by the number of items taken out from the shelf **112**.

For example, when the weight change is an increase in weight, the information processing unit **1544** reads the merchandise code stored in association with the shelf ID of the shelf **112** having the weight change. Then, the information processing unit **1544** refers to the merchandise master **1622** and identifies the merchandise code corresponding to the increased weight among the weights associated with the read merchandise codes, thereby identifying the merchandise returned to the shelf **112**. In this case, the information processing unit **1544** deletes the merchandise information corresponding to the identified merchandise code registered in the RAM **153**. The information processing unit **1544** increases the quantity of the identified merchandise codes stored in the accommodated merchandise table **1623** by the number of items returned to the shelf **112**.

The registration of the merchandise information may be executed by the server device **30**. Specifically, the information processing unit **1544** transmits the merchandise code of the merchandise taken out from the shelf **112** to the server device **30** in the same manner as described above. Then, the server device **30** reads the merchandise information on the merchandise corresponding to the received merchandise code from the merchandise master **3162** and registers the merchandise information. In this case, the information processing unit **1544** has a function of causing the server device **30** to register the merchandise information, and can be said to function as the registration unit that executes the merchandise registration.

When the input reception unit **1542** receives the settlement instruction, the information processing unit **1544** executes the payment. The payment is a part or all of a process for the user to pay a purchase price of the merchandise, and includes calculation of the purchase price of the merchandise subjected to the merchandise registration, acquisition of the face image data, and transmission of the settlement request in the present embodiment.

When the mobile sales robot **10** executes the settlement executed by the server device **30**, the payment executed by the information processing unit **1544** includes the settlement. The mobile sales robot **10** may include an automatic coin machine or the like for cash settlement. In this case, the information processing unit **1544** controls the automatic coin machine. In this way, the information processing unit **1544** functions as a payment unit that executes the payment for part or all of the process related to the payment of the purchase price.

The information processing unit **1544** executes sound recognition on the sound collected by the sound collection unit **157** from the periphery, and outputs a stop instruction to the drive control unit **1545** as necessary. For example, when the information processing unit **1544** recognizes a sound for stopping the mobile sales robot **10** by the sound recognition, the information processing unit **1544** outputs the stop instruction to the drive control unit **1545**.

When the transmission and reception unit **1541** receives the authentication notification from the server device **30**, the information processing unit **1544** permits the process related to the sales of the merchandise. That is, when the user who called the mobile sales robot **10** and the user who now intends to start shopping are determined to match with each other by the collation unit **3143** of the server device **30**, the information processing unit **1544** permits the process related to the sales of the merchandise.

For example, the information processing unit **1544** cooperates with the display control unit **1546** to display an initial screen for starting shopping on the display unit **155**, thereby permitting the process related to the sales of the merchandise. When a lockable door is provided in the accommodation portion **111**, the information processing unit **1544** permits the process related to the sales of the merchandise by releasing a lock of the door. When the user who called the mobile sales robot **10** and the user who intends to start shopping match, the information processing unit **1544** functions as a permission unit that permits the process related to the sales of the merchandise.

In the present embodiment, the trigger information for permitting sales of the merchandise at the call position is the authentication notification from the server device **30**. However, the trigger information is not set during the circulate traveling of the mobile sales robot **10**. This is because, during the circulate traveling of the mobile sales robot **10**, the mobile sales robot **10** stops at a sales position based on the sound of the user, that is, in this case, the call request and the pin code corresponding to the call request are not issued. For the sales of the merchandise during the circulate traveling of the mobile sales robot **10**, the trigger information for permitting the process related to the sales of the merchandise may not be set, or other information different from the authentication notification may be used as the trigger information.

For example, the trigger information may be detection that the user makes eye contact with the imaging unit **134**, detection that the user performs a predetermined action such as blinking in front of the imaging unit **134**, detection that the user operates the disinfection device **135**, or the like. The user who purchases the merchandise during the circulate traveling of the mobile sales robot **10** can purchase the merchandise by the settlement based on the face authentication when the settlement information is registered in the user management table **3165** in advance. When the mobile sales robot **10** includes a settlement device such as a card settlement terminal or a coin machine, a user can purchase merchandise regardless of whether the settlement information is registered.

For the sales of the merchandise at the designated call position, the trigger information for permitting the processes related to the sales of the merchandise may be other information instead of the authentication notification, or may be information simply indicating that the mobile sales robot **10** arrives at the call position. Specific trigger information for permitting the processes related to the sales of the merchandise may not be required.

The information processing unit **1544** executes alarming for issuing an alarm as necessary. For example, when it is detected that the merchandise is illegally taken out from the shelf **112** while the mobile sales robot **10** is traveling or in a state in which the process related to the sales of the merchandise is not permitted, the information processing unit **1544** executes the alarming. The alarming causes the light emitting unit **114** to emit light in order to notify the surroundings of an abnormality. The color is changed to a color indicating a warning, and an alarm sound is generated from a speaker or the like.

The information processing unit **1544** may execute the alarming when it is detected that the merchandise is about to be taken out illegally. For example, when it is detected that a hand has been inserted into the accommodation portion **111** while the mobile sales robot **10** is traveling or the processing related to the sales of the merchandise is not presently permitted, the alarming may be executed. The information processing unit **1544** functions as an alarm unit that issues an alarm when it is detected that merchandise taken out illegally or is about to be taken out illegally.

The drive control unit **1545** controls the drive unit **123** to cause the mobile sales robot **10** to travel or stop. Under the control of the drive control unit **1545**, the mobile sales robot **10** performs the circulate traveling along the predetermined route and the call traveling toward the call position designated by the customer.

The drive control unit **1545** moves the mobile sales robot **10** to, for example, a maintenance yard (storage room) when it is detected that any one item or all items accommodated in the accommodation portion **111** are out of stock. This is to replenish the merchandise out of stock. When the mobile sales robot **10** travels in a shopping mall or the like, the drive control unit **1545** may move the mobile sales robot **10** to a store that sells the merchandise out of stock.

The display control unit **1546** controls the display unit **155** to display various types of information on the display unit **155**. The display control unit **1546** causes the first display unit **131**, the second display unit **132**, and the third display unit **133** to display contents corresponding to the arrangement positions. For example, the first display unit **131** displays information related to the merchandise taken out from the accommodation portion **111**. The second display unit **132** displays information related to sales promotion of the merchandise, such as information for advertising the merchandise accommodated in the accommodation portion **111**. The third display unit **133** displays information related to the settlement of the merchandise, such as information for guiding a merchandise settlement method. When the shelf label **113** is an electronic shelf label, the display control unit **1546** also controls the display of the electronic shelf label.

An operation of the mobile sales system **1** including the devices having the above-described configurations will be described. FIG. **24** is a sequence chart illustrating an example of the operation of the mobile sales system **1**. The sequence chart illustrates a case in which the user checks the merchandise information on the merchandise mounted on the mobile sales robot **10**, calls the mobile sales robot **10**, and purchases the merchandise by the mobile sales robot **10**.

When calling the mobile sales robot **10**, the user activates an application program of the mobile sales system **1** provided in the user terminal **20**. When the merchandise information request menu is selected, the user terminal **20** displays the first display screen illustrated in FIG. **16** and receives the selection of the mobile sales robot **10** (Act **1**). The user terminal **20** transmits a merchandise information request including the robot ID of the selected mobile sales robot **10** and the terminal ID of the user terminal **20** to the server device **30** (Act **2**).

The server device **30** reads the merchandise information from the storage unit **316** according to the received merchandise information request (Act **3**). Specifically, the server device **30** reads a merchandise code corresponding to the robot ID included in the merchandise information request from the merchandise management table **3164**. Next, the server device **30** reads merchandise information corresponding to the read merchandise code from the merchandise master **3162**. The server device **30** transmits the read merchandise information to the user terminal **20** (Act **4**). A communication address of the user terminal **20** may be registered in the user management table **3165** in advance or may be included in the merchandise information request.

The user terminal **20** displays the received merchandise information on the display unit **215** (Act **5**). When the user checks the displayed merchandise information and there is a merchandise to be purchased, the user performs an operation for calling the mobile sales robot on which the merchandise is mounted. Accordingly, a call request is input to the user terminal **20** (Act **6**). The user terminal **20** acquires position information of a place where the user terminal **20** is present (Act **7**), and transmits a call request in which the acquired position information is set as a call position to the server device **30** (Act **8**).

Upon receiving the call request, the server device **30** executes call request reception (Act **9**). The call request reception will be described in detail later. The server device **30** transmits a pin code issued in the call request reception to the user terminal **20** (Act **10**). The server device **30** transmits a movement instruction including the call position to the mobile sales robot **10** designated by the call request (Act **11**).

The mobile sales robot **10** controls the drive unit **123** to move to the call position designated by the movement instruction (Act **12**). When the mobile sales robot **10** arrives at the call position, the mobile sales robot **10** transmits an arrival notification to the server device **30** (Act **13**). The mobile sales robot **10** may transmit only the position information indicating the position where the mobile sales robot **10** is present to the server device **30**. The server device **30** may determine the arrival of the mobile sales robot **10** at the call position.

The server device **30** updates the call management table **3166** based on the received arrival notification (Act **14**). Specifically, the server device **30** registers an arrival date and time of the call request in the call management table **3166**. Next, the server device **30** transmits an arrival notification to the user terminal **20** which is the request source of the call request (Act **15**).

When the user terminal **20** receives the arrival notification, the user terminal **20** displays, on the display unit **215**, information indicating that the called mobile sales robot **10** arrives (Act **16**). Accordingly, the user recognizes the arrival of the mobile sales robot **10** and inputs the pin code received from the server device **30** to the operation unit **156** of the mobile sales robot **10**.

When the mobile sales robot **10** receives the input of the pin code (Act **17**), the mobile sales robot **10** transmits an authentication request including the input pin code and robot ID to the server device **30** (Act **18**).

The server device **30** authenticates the pin code (Act **19**). Specifically, when a combination of the pin code and the robot ID included in the received authentication request is stored in the call management table **3166**, the server device **30** authenticates the pin code. The server device **30** authenticates the pin code, and transmits an authentication notification to the mobile sales robot **10** when it is recognized that the user who issues the call request and the user who is about to start shopping match with each other (Act **20**).

When the mobile sales robot **10** receives the authentication notification, the mobile sales robot **10** permits a process related to the sales of the merchandise. Then, when it is detected that the merchandise is taken out from the accommodation portion **111** (Act **21**), the merchandise registration is executed for the merchandise (Act **22**). The mobile sales robot **10** executes the merchandise registration every time the merchandise is taken out from the shelf **112**, and cancels the merchandise registration when the merchandise is returned to the shelf **112**. When a payment instruction is input, the mobile sales robot **10** executes payment processing (Act **23**). Then, the mobile sales robot **10** transmits a settlement request to the server device **30** based on the received payment (Act **24**).

The server device **30** executes settlement processing based on the received settlement request (Act **25**). When the settlement is normally completed, the server device **30** transmits a settlement completion notification to the mobile sales robot **10** (Act **26**) and also transmits the settlement completion notification to the user terminal **20** (Act **27**). When the user terminal **20** receives the settlement completion notification, the user terminal **20** displays, on the display unit **215**, information indicating that the settlement is completed (Act **28**).

By the above operation, the mobile sales system **1** can move the mobile sales robot **10** on which the merchandise is mounted to the position designated by the user to sell the merchandise.

Next, the call request reception executed by the server device **30** will be described. FIG. **25** is a flowchart illustrating an example of a flow of the call request reception executed by the control unit **314** of the server device **30**.

The control unit **314** determines whether the transmission and reception unit **3141** receives a call request from the user terminal **20** (Act **31**). When the transmission and reception unit **3141** does not receive the call request (N in Act **31**), the process returns to Act **31** and waits. When the transmission and reception unit **3141** receives the call request (Y in Act **31**), the control unit **314** issues a pin code associated with the call request (Act **32**).

The information management unit **3142** registers information related to the call request received by the transmission and reception unit **3141** in the call management table **3166** (Act **33**). Next, the transmission and reception unit **3141** transmits the issued pin code to the user terminal **20** which is the request source of the call request (Act **34**).

The robot management unit **3145** calculates an arrival time until the mobile sales robot **10** designated by the call request arrives at the call position, and transmits the calculated arrival time to the user terminal **20** (Act **35**). The robot management unit **3145** refers to the robot management table **3163** and calculates the arrival time based on the position of the mobile sales robot **10** and the call position when the state of the designated mobile sales robot **10** is "waiting for call" capable of responding to the call from the user.

When the state of the designated mobile sales robot **10** is not "waiting for call", the robot management unit **3145** calculates the arrival time in consideration of other information. Here, the other information is, for example, the number of call requests (still waiting requests whose arrival date and time are not registered) to the designated mobile sales robot **10** registered in the call management table **3166**, and a purchase time limit in one transaction set in advance.

The robot management unit **3145** determines whether the state of the mobile sales robot **10** designated by the call request is "waiting for call" (Act **36**). When the mobile sales robot **10** is "waiting for call" (Y in Act **36**), the transmission and reception unit **3141** transmits a movement instruction including the terminal ID and the call position of the user terminal **20** to the designated mobile sales robot **10** (Act **37**). Then, the control unit **314** ends the call request reception.

When the state of the designated mobile sales robot **10** is not "waiting for call" (N in Act **36**), that is, when the mobile sales robot **10** cannot immediately start moving to the call position, the control unit **314** returns to Act **36**.

Through the call request reception, the server device **30** can manage the call requests from the user terminals **20** and move the designated mobile sales robot **10** to the call position in the order in which the call requests are received.

Next, the payment executed by the mobile sales robot **10** will be described. FIG. **26** is a flowchart illustrating an example of a flow of the payment executed by the control unit **154** of the mobile sales robot.

The control unit **154** determines whether the input reception unit **1542** receives a settlement instruction (Act **41**). When the input reception unit **1542** does not receive the settlement instruction (N in Act **41**), the process returns to Act **41** and waits. When the input reception unit **1542** receives the settlement instruction (Y in Act **41**), the information processing unit **1544** reads the merchandise information on the merchandise subjected to the merchandise registration (Act **42**).

The display control unit **1546** causes the first display unit **131** to display the merchandise information read by the information processing unit **1544** (Act **43**). The information processing unit **1544** calculates, based on the read merchandise information, a transaction amount, that is, a total amount of prices of items subjected to the merchandise registration (Act **44**). The calculated transaction amount is temporarily stored in the RAM **153**.

The control unit **154** determines whether the information processing unit **1544** acquires face image data from the imaging unit **134** (Act **45**). When the information processing unit **1544** does not acquire the face image data (N in Act **45**), the process returns to Act **45** and waits. In other words, the control unit **154** determines whether the imaging unit **134** captures an image of the face of the user who purchases the merchandise in order to perform the settlement by face authentication.

When the information processing unit **1544** acquires the face image data from the imaging unit **134** (Y in Act **45**), the transmission and reception unit **1541** transmits a settlement request to the server device **30** (Act **46**). The settlement request includes the face image data, the merchandise information, the transaction amount, and the like. Then, the control unit **154** ends the payment.

Through the payment, the mobile sales robot **10** can provide the server device **30** with information necessary for the server device **30** to perform the settlement, and can complete the payment of the purchase price of the user in cooperation with the server device **30**.

Next, the settlement executed by the server device will be described. FIG. **27** is a flowchart illustrating an example of a flow of the settlement by the control unit **314** of the server device.

The control unit **314** determines whether the transmission and reception unit **3141** receives a settlement request from the mobile sales robot **10** (Act **51**). When the transmission and reception unit **3141** does not receive the settlement request (N in Act **51**), the process returns to Act **51** and waits.

When the transmission and reception unit **3141** receives the settlement request (Y in Act **51**), the settlement unit **3144** determines whether the user ID can be identified by the face image data included in the settlement request (Act **52**). Specifically, the settlement unit **3144** refers to the user management table **3165** and searches whether the received face image data or the feature data of the face image data is registered as the feature information.

When the user ID can be identified (Y in Act **52**), the settlement unit **3144** reads the settlement information from the user management table **3165** and transmits the settlement information to a settlement server or the like of the settlement business operator (Act **53**). Subsequently, the settlement unit **3144** determines whether the settlement has been completed based on whether information is received from the settlement server (Act **54**).

When the settlement is completed (Y in Act **54**), the settlement unit **3144** updates the transaction management table **3167** (Act **55**). Specifically, the settlement unit **3144** registers information indicating settlement completion in the settlement flag of the transaction management table **3167**. Next, the transmission and reception unit **3141** transmits a transaction completion notification to the user terminal **20** and the mobile sales robot **10** (Act **56**). The transmission and reception unit **3141** selects a terminal ID corresponding to an arrival date and time closest to a time point at which the settlement request is received among the terminal IDs corresponding to the robot IDs of the mobile sales robots **10** on which the items are sold in the call management table **3166**, and transmits the settlement completion notification to the user terminal **20** having the terminal ID. Then, the control unit **314** ends the settlement.

In Act **52**, when the user ID cannot be identified (N in Act **52**), that is, when the face image data included in the settlement request or the feature data of the face image data is not registered in the user management table **3165**, the transmission and reception unit **3141** transmits error information indicating that the settlement cannot be performed to the mobile sales robot **10** (Act **57**). Then, the control unit **314** ends the settlement.

When the settlement is not completed (N in Act **54**), the control unit **314** proceeds to Act **57**. When the settlement performed by the settlement business operator can not be completed, the transmission and reception unit **3141** transmits error information to the mobile sales robot **10**.

The server device **30** performs the settlement processing in response to the settlement request from the mobile sales robot **10**. Accordingly, the user can pay the purchase price without having to go to a register counter or the like separately.

As described above, a mobile sales system **1** includes a mobile sales robot **10** with an accommodation portion **111** configured to accommodate items to be sold and a drive unit **123** configured to move the mobile sales robot **10**. A user terminal **20** is configured to designate a call position for the mobile sales robot **10**. The user terminal **20** includes a transmission and reception unit **2141** configured to transmit position information indicating the call position for the mobile sales robot **10**. The mobile sales robot **10** includes a transmission and reception unit **1541** configured to acquire the position information transmitted by the user terminal **20**, a drive control unit **1545** configured to control the drive unit **123** based on the position information acquired by the transmission and reception unit **1541**, and a payment unit (e.g., an information processing unit **1544**) configured to execute payment processing based on merchandise information for any merchandise removed from the accommodation portion **111**.

Accordingly, the mobile sales system **1** can move a mobile sales robot **10** on which merchandise is mounted to a position designated by a user and sell the merchandise to the user. A business operator who sells merchandise can broaden sales ranges by adopting the mobile sales system **1**.

The user terminal **20** in the mobile sales system **1** according to an embodiment is a terminal device that can be carried by a user, and further includes the self-position identifying unit **2143** configured to identify a self-position (present device location). The transmission and reception unit **2141** is configured to transmit position information indicating a position identified by the self-position identifying unit **2143** as the position information indicating the call position.

Accordingly, the user can call the mobile sales robot **10** without separately performing an operation for inputting the call position to the mobile sales robot **10**. Therefore, the mobile sales system **1** can improve user convenience.

The user terminal **20** in the mobile sales system **1** according to an embodiment includes the transmission and reception unit **2141** configured to receive merchandise information for the merchandise mounted on the mobile sales robot **10**, and the display control unit **2144** configured to cause the display unit **215** to display the merchandise information received by the transmission and reception unit **2141**.

Accordingly, the user can call the mobile sales robot **10** after checking what merchandise is on the mobile sales robot **10**. Therefore, the mobile sales system **1** can improve the user convenience in this respect as well.

In addition, the mobile sales robot **10** in the mobile sales system **1** according to an embodiment includes a detection unit (e.g., an information processing unit **1544**) configured to detect merchandise taken out from the accommodation portion **111**. The payment unit (e.g., an information processing unit **1544**) executes the payment processing for the merchandise detected by the detection unit.

Accordingly, the user does not need to perform an operation for separately reading (e.g., scanning) the merchandise information on the merchandise to be purchased. Therefore, the user can more easily shop and pay the purchase price.

The mobile sales robot **10** in the mobile sales system **1** according to an embodiment includes an input unit (e.g., a transmission and reception unit **1541**) to which trigger information (e.g., an authentication notification) serving as a condition to start sales can be input, and an alarm unit (e.g., an information processing unit **1544**) configured to issue an alarm when it is detected that an item in the accommodation portion **111** has been taken out before the input unit has received the trigger information.

Accordingly, the mobile sales robot **10** can issue an alarm when an item is illegally taken out from the accommodation portion **111** or when it seems the item is about to be taken out. Therefore, security of the mobile sales system **1** can be improved.

In an embodiment, control programs executed by the mobile sales robot **10**, the user terminal **20**, and the server device **30** may be provided by being recorded in a computer-readable recording medium such as a CD-ROM. The control programs executed by each of the devices may be stored in a computer connected to a network such as the Internet and provided by being downloaded via the network, or may be accessed via the network such as the Internet.

While the embodiment is described above, the embodiment is presented as an example only, and is not intended to limit the scope of the disclosure. The above-described embodiment can be implemented in various other forms, and various omissions, substitutions, and changes can be made without departing from the spirit of the disclosure. The embodiment and a modification thereof are included in the scope and gist of the disclosure, and are included in the scope of the disclosure disclosed in the scope of claims and equivalent scope thereof.

What is claimed is:

1. A mobile sales system, comprising:
- a mobile sales device on which items to be sold can be mounted; and
- a terminal device to designate a customer call position for the mobile sales device, wherein
- the terminal device is configured to:
  - transmit position information indicating the customer call position, and
- the mobile sales device is configured to:
  - acquire the position information transmitted by the terminal device,
  - move to a location corresponding to the acquired position information,
  - transmit an arrival notification upon arriving at the location,
  - require an input of authentication information before permitting a sales transaction start,
  - issue an alarm if any item is removed from the mobile sales device before the authentication information has been input, and
  - execute payment processing for an item removed from the mobile sales device by a customer, wherein
- the terminal device is further configured to display a sales transaction start time limit screen indicating an elapse of a predetermined period of time from the transmitting of the arrival notification by the mobile sales device.

2. The mobile sales system according to claim **1**, wherein the terminal device is a mobile device carried by the customer.

3. The mobile sales system according to claim **2**, wherein
- the terminal device comprises a self-location unit configured to identify a current position of the terminal device, and
- the position information transmitted by the terminal device is the current position of the terminal device.

4. The mobile sales system according to claim **1**, wherein the terminal device is further configured to:
- receive merchandise information about items mounted on the mobile sales device, and
- cause a display unit of the terminal device to display the received merchandise information.

5. The mobile sales system according to claim **1**, wherein the mobile sales device is further configured to:
- detect each item removed from the mobile sales device, and
- receive payment for each item detected as removed from the mobile sales device.

6. The mobile sales system according to claim **1**, wherein the mobile sales device is a wheeled robot.

7. The mobile sales system according to claim **1**, wherein the mobile sales device includes:
- a merchandise display portion on which items are mounted, and
- a moving unit below the merchandise display portion which moves the mobile sales device from place to place.

8. The mobile sales system according to claim **1**, further comprising:
- a server configured to connect to the mobile sales device and the terminal device via a network, wherein the server is configured to:
- receive a selection of one of a plurality of mobile sales devices from the terminal device along with the position information indicating the call position,
- transmit the position information to the selected one of the plurality of mobile sales devices, and
- receive the arrival notification from the mobile sales device and transmit a corresponding notification to the terminal device to cause the terminal device to display the sales transaction start time limit screen.

9. A mobile sales system, comprising:
- a plurality of mobile robots on which items to be sold can be mounted;
- a server device connected to the plurality of mobile robots via a network connection;
- a terminal device to designate a customer call position for a mobile robot of the plurality of mobile robots, wherein
- the terminal device is configured to:
  - transmit position information indicating the customer call position to the server device, and
- each mobile robot is configured to:
  - acquire the position information transmitted by the terminal device to the server,
  - move to a location corresponding to the acquired position information,
  - transmit an arrival notification upon arriving at the location to the server device,
  - require an input of authentication information before permitting a sales transaction start,
  - issue an alarm if any item is removed from the mobile robot before the authentication information has been input, and
  - execute payment processing for an item removed from the mobile robot by a customer, wherein
- the terminal device is further configured to display a sales transaction start time limit screen indicating an elapse of a predetermined period of time from the transmitting of the arrival notification by the mobile robot based on a notification of arrival from the server device.

10. The mobile sales system according to claim **9**, wherein the terminal device is further configured to:
- receive merchandise information about items mounted on each of the mobile robots in the plurality of mobile robots, and
- cause a display unit of the terminal device to display the received merchandise information.

11. The mobile sales system according to claim **9**, wherein each mobile robot is further configured to:
- detect any item being removed from the mobile robot, and
- receive payment for each item detected as removed from the mobile robot.

12. The mobile sales system according to claim 9, wherein each mobile robot is a wheeled robot.

13. The mobile sales system according to claim 9, wherein each mobile robot includes:

a merchandise display portion on which items are mounted, and

a moving unit below the merchandise display portion which moves the mobile robot from place to place.

14. A server device for use in a mobile sales system comprising a mobile sales device on which items to be sold can be mounted and a terminal device configured to designate a customer call position, the server device comprising:

a communication interface to connect to a terminal device and a mobile sales device; and

a processor configured to:

receive a selection indicating a mobile sales device and position information indicating the customer call position for the mobile sales device,

transmit the position information received from the terminal device to the selected mobile sales device,

receive an arrival notification from the selected mobile sales device indicating an arrival of the selected mobile sales device at a location corresponding to the customer call position, and

upon receiving the arrival notification, send a notification of arrival to the terminal device that causes the terminal device to display a sales transaction start time limit screen indicating an elapse of predetermined period of time from the transmitting of the arrival notification by the selected mobile sales device.

15. The server device according to claim 14, wherein the processor is further configured to send authentication information to the terminal device after receiving the selection indicating the mobile sales device and the position information.

16. The server device according to claim 14, wherein the processor is further configured to send merchandise information indicating items mounted on the mobile sales device to the terminal device.

17. The server device according to claim 14, wherein the processor is further configured to perform settlement processing for any items removed from the mobile sales device.

18. The mobile sales system according to claim 1, wherein the mobile sales device is further configured to cancel a sales transaction request at the location upon the elapse of the predetermined period of time.

19. The mobile sales system according to claim 9, wherein each mobile robot is further configured to cancel a sales transaction request at the location upon the elapse of the predetermined period of time.

20. The server device according to claim 14, wherein the processor is further configured to cancel a sales transaction request at the location upon the elapse of the predetermined period of time.

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